

PhyReAct: Agentic Physical Reasoning for Video Evaluation and Refinement

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Visual fluency in generated video does not imply physical reliability, and a scalar quality score names neither the obligation a clip violates nor the moment it fails. We present **PhyReAct**, a physically grounded instance of the ReAct agentic paradigm: a reasoning model runs a thought–action–observation loop in which reasoning compiles the prompt into typed physical obligations, and observations gate and scope which operators fire next. The actions invoke frozen low-level experts—segmentation and tracking, counting, eleven typed physical measurements over the resulting tracks, depth, OCR, and audio event detection—that return measurements. A deterministic layer composes those measurements into a three-valued state (**supported**, **contradicted**, or **unknown**, surfaced as *plausible*, *implausible*, or **abstain**) with full provenance, so that every verdict is traceable to the measurement that produced it. We anchor evaluation in a 1,500-clip corpus of human-annotated flaw records that localize real generation failures in prompt reference, space, and time. On a 149-clip core carrying 304 such records, **PhyReAct** accounts for 228, against 164 for a published question-decomposition evaluator given the same clips and the same claims. Recall alone does not separate it from prompting the same backbone monolithically, which reaches 222; what separates them is that each of **PhyReAct**’s 228 arrives with the measurement that forced it, making the traces auditable one verdict at a time and usable as the interface through which a critic verdict could be written back into generation. The core is a development set, so these figures characterize the system rather than its generalization.

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1 Introduction

Text-to-video systems now produce clips with striking appearance and smooth local motion, yet visual fluency is not physical reliability. A generated video can preserve style and identity while omitting a requested object, reversing the order of two events, violating a contact outcome, or following a trajectory inconsistent with the described scene. Recent benchmarks expose this gap between perceptual quality and physical faithfulness [3, 4, 44, 48]. For evaluation that could eventually drive correction, however, a scalar quality score is not enough: it cannot say which prompt-conditioned obligation failed—over entities, events, motion, contact, space, quantity, and time—where in the clip it failed, or on what evidence the judgment rests. What is missing is not a better number but a decision that names the violated obligation and exposes the evidence behind it.

We approach physical evaluation as physically grounded reasoning in the reason–act–observe style. ReAct established the general pattern of interleaving a model’s reasoning with tool actions and observations of their results [77]. **PhyReAct** is a grounded instance of that pattern specialized to physical verification: a reasoner compiles the prompt into *typed physical obligations* and an executable measurement plan, and observations returned by physical operators then gate and scope which later operator calls are valid and where in the video they run. The system belongs to the plan-then-execute branch of that family: reasoning declares the obligations and their measurement plan up front, and observations activate, skip, or localize measurements the plan already contains. Run-time freedom is confined to that gating and scoping.

Physical grounding is what specializes the loop. Frozen operators — grounding and tracking with SAM 3, counting over its instance identities, eleven typed physical measurements over those tracks, monocular depth, optical character recognition, and audio event detection — return measurements; a reasoning model supplies



vs.

PhyReAct (ours) plan, then measure on the timeline

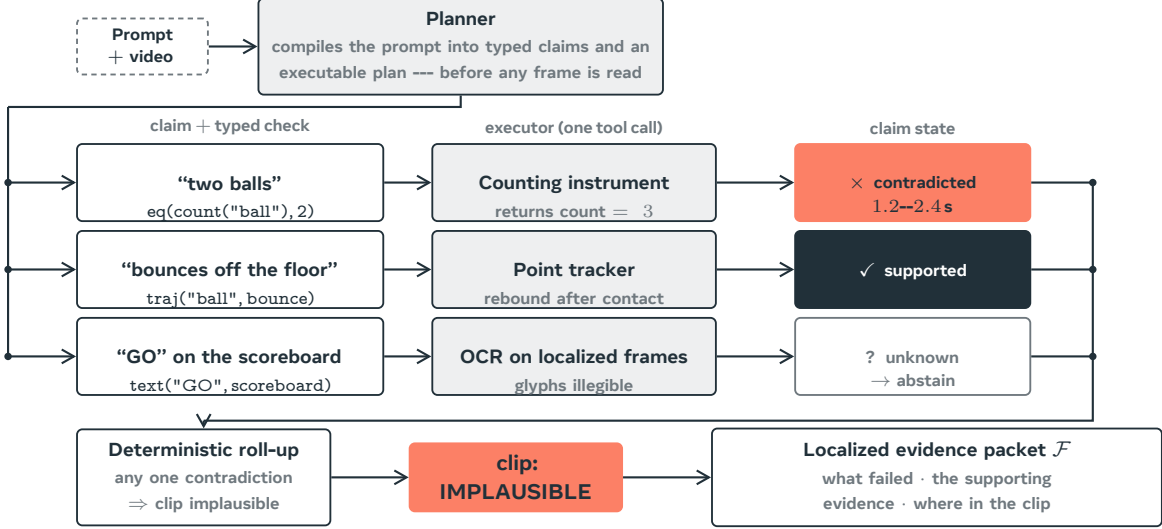


Figure 1 PhyReAct turns an opaque quality score into an auditable verdict. A learned video-text scorer maps a prompt and clip to a single number that cannot say which prompt obligation failed, where in the clip, or on what evidence. PhyReAct instead runs a reason-act-observe loop: reasoning compiles the prompt into typed physical obligations and an executable measurement plan, frozen physical operators return localized measurements, and fixed rules combine them into **supported**, **contradicted**, or **unknown** claim states, a *plausible*, *implausible*, or **abstain** clip decision, and a localized evidence packet recording what failed, when, and on what basis. The claims and windows shown are an illustrative schematic of the output contract, not a measured run.

the semantics; and deterministic rules compose the measurements into a three-valued decision carrying full provenance.

Four roles divide the labor. A *text-only planner* reads the whole prompt and writes the typed obligations and their measurement plan before any frame is seen. A *video-aware semantic verifier* reads timestamped frames densely for predicates such as existence and event occurrence. *Physical operators* supply targeted measurements—counts, tracks, masks, depths, rendered text, and sound intervals. A *deterministic aggregator* then applies typed predicates and three-valued composition, so that no learned component emits the final label. The video’s timeline structures execution as a conditioned evidence chain: an event must be observed before its time interval can support anything else (occurrence gates timing), that confirmed interval restricts the frames a more expensive operator examines (localized scope), and temporal and spatial relations are then computed from the resulting intervals by fixed rules. Earlier observations therefore determine which later measurements are valid and where they are taken, rather than leaving evidence as an unordered bag of model answers. The output is a claim-level trace that a person can read, audit, and hand off to human-in-the-loop review (figure 1).

Evaluating such a critic requires ground truth at the same level of detail. We therefore assemble a corpus of 1,500 generated clips with 2,582 human-written flaw records, each quoting the exact prompt fragment that was violated and adding a rationale, a severity, a confidence, and, where the annotator could supply them, a temporal span and an object track. These records are the only ground truth in this work, and because each names a particular failure, an allegation the critic makes can be matched to the one a person marked.

The same structured trace also defines an interface to refinement: a contradicted claim names what should

change, and its localized evidence indicates when and where an intervention is needed. As a controllable testbed for this direction, we render MuJoCo simulations as depth or silhouette control videos for a Wan 2.2-VACE generator, so that geometry and motion come from the simulator while appearance comes from the text prompt [29, 64, 65]. This gives a control channel through which critic feedback could later update *how a clip is generated*. In this report the critic and the generator are studied separately; closing the critic-to-generator loop remains future work.

Reading physical evaluation as a reason–act–observe framework also makes several extensions natural, which we state as design directions. Because the plan is written as a program over operators, the operator library can grow with experience into a reusable, cost-annotated toolset [50, 73]. Because every trace is inspectable and version-controllable, traces and their feedback could accumulate as human-readable prior context and let the system evolve without updating weights [27, 82]. And because a loop that feeds only its own traces back into its own context is internally closed, admitting external or social knowledge would break that closure—but only under a candidate-and-pruning discipline that guards against context pollution [27]. These are directions the framework opens.

We make three contributions. First, **PhyReAct**, a physically grounded reason–act–observe critic that compiles a prompt into typed physical obligations, gates and scopes frozen physical operators over the video timeline, and composes their measurements into a three-valued verdict in which every conclusion is attributable to the measurement that forced it. Second, a 1,500-clip corpus of human-annotated, prompt-grounded flaw records, together with the flaw-level protocol that makes a critic and a published evaluator comparable at all: each defect is judged whole, partial, or missed against what an annotator wrote, so finding one aspect of a compound complaint is neither credited as the whole nor discarded as nothing. Third, a simulation-conditioned generation testbed in which known MuJoCo geometry drives a frozen video backbone, giving a control channel of the exact shape a localized critic verdict could later be written into.

On the flaw-level protocol, **PhyReAct** accounts for 228 of the 304 annotated defects; a published question-decomposition evaluator, given the same clips, the same claims, and the same served model, accounts for 164. Choosing what to measure, and measuring it, reaches defects that putting a yes-or-no question to each claim does not. Recall does not separate **PhyReAct** from prompting the same backbone monolithically, which reaches 222; the evidence behind each verdict does, and [section 5](#) reports both figures with the protocol and its limits. These 149 clips are recall-only, single-annotator, and not held out, so they characterize the system as it stands rather than its generalization.

2 Physics-Guided Video Generation

A text-to-video model decides how a scene looks and how it moves from a single text input, and the motion it produces frequently violates physical constraints. Benchmarks built to test exactly this report systematic failures of physical commonsense in current models [3, 4, 48], Methods that revise the prompt and regenerate act through that same input [72]: one text description specifies appearance and dynamics together, so the resulting dynamics are those the model infers from that description.

Appearance and motion are separable, and the separation can be measured. Holding a geometric control video fixed while varying only the motion clause of the prompt, the clause alone moves the generated object’s horizontal travel across almost the whole frame when no control is applied, and moves it not at all when the control is applied throughout sampling ([section 2.4](#)). With the control present the prompt’s motion clause has no measurable effect on the trajectory, including a clause that specifies the opposite motion.

The control also need not act over the whole of sampling. The reverse diffusion process fixes large-scale structure before fine detail [10, 24, 25, 53], and prior theory describes a critical window — a relatively high-noise interval at which the trajectory commits to a global structure, after which later steps refine within it [5, 34, 52, 56]. The mapping from spatial coarse-to-fine structure to temporal dynamics is unestablished, and results in other temporal domains do not exhibit it [13]. We therefore call a trajectory *committed* at a given noise level when withdrawing the geometric conditioning at that level leaves the generated motion unchanged, and measure that level directly ([section 2.5](#)).

This section builds a generator on that arrangement and measures both halves of it. A physics description is

Bouncing ball

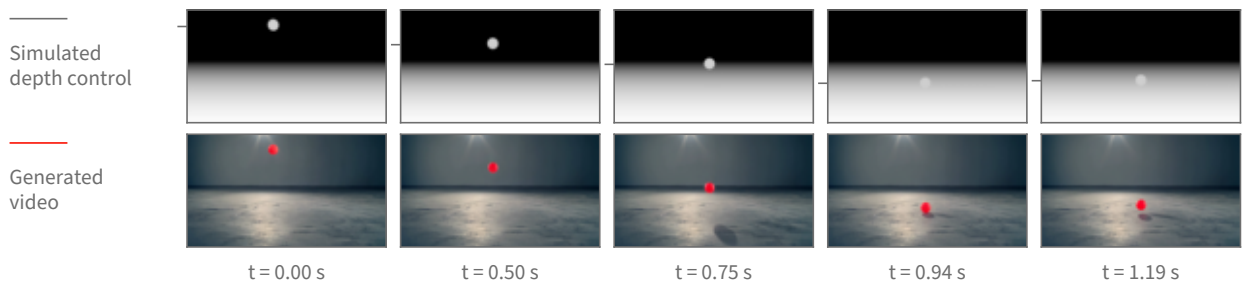


Figure 2 Physics-guided generation on one bouncing-ball clip. **Top:** the depth control video rendered from the rigid-body simulation, which is the backbone’s geometric input. **Bottom:** the clip generated from it, with appearance from the text prompt. Both rows are sampled at the same five timestamps of the same 5.06 s, 16 fps, 848×480 clip, placed at the turning points of the ball’s vertical path so the columns show the drop, the floor contact and the rebound. Tracked independently in each row by the colour tracker of [section 2.5](#), the two paths agree to a correlation of 1.0000 horizontally and 0.9997 vertically over all 81 frames.

decomposed into a scene specification, simulated, and rendered into a control video that a diffusion backbone follows for the early, coarse portion of sampling, while the text prompt alone supplies appearance throughout ([figure 2](#)). We then measure what each input determines, locate the noise level at which the motion stops responding to removal of the control, and show which component of the motion is the last to become fixed.

2.1 From a physics description to a control video

The scene specification is produced from the text description by a language model and accepted only once simulation confirms that it realizes the described behaviour ([figure 3](#)).

Given the physical event the clip is required to show, the model emits a structured scene specification: the bodies, their initial positions and velocities, the static surfaces they interact with, and a camera. Each body is either given to the engine with an initial velocity and its colliders, so that its motion is computed, or given prescribed waypoints. A body on a ballistic segment takes the first form, and the simulator computes the arc, the contact and the subsequent rest. Waypoints are reserved for motion the engine would otherwise override: a self-propelled body moving under its own power, or one that must contradict the ordinary physical prior, such as a rising balloon.

Validation is applied to the simulated trajectory. A ballistic segment must trace an arc that reaches its target, or for a counterfactual cleanly misses it, and comes to rest; a rising segment must gain height; no body may leave the frame. A specification failing any condition is returned to the model with the violated condition attached, together with a summary of the motion the engine actually produced, and re-authored. The loop runs for at most three rounds, after which the latest specification is carried forward whether or not it passes. Acceptance is therefore a property of the simulated trajectory: a specification is accepted when the trajectory it produces meets every condition, and proximity to the target without a ballistic path does not.

The accepted specification yields a simulation whose dynamics are computed by the integrator, and is rendered into a silhouette control of the kind described in [section 2.3](#). This authoring route runs a planar rigid-body engine; the depth controls of the scene library are produced by the three-dimensional engine of [section 2.3](#). In both cases the control video is rendered from a solver state.

2.2 Controlling motion and appearance separately

The generator takes four inputs. Geometry and motion are carried by a control video and appearance by the text prompt; the remaining two govern how the first is applied, a mask stating which pixels are synthesised and a scalar conditioning strength setting how strongly the control acts.

The backbone that consumes them is Wan 2.2-VACE-Fun-14B, a two-expert mixture-of-experts video-diffusion

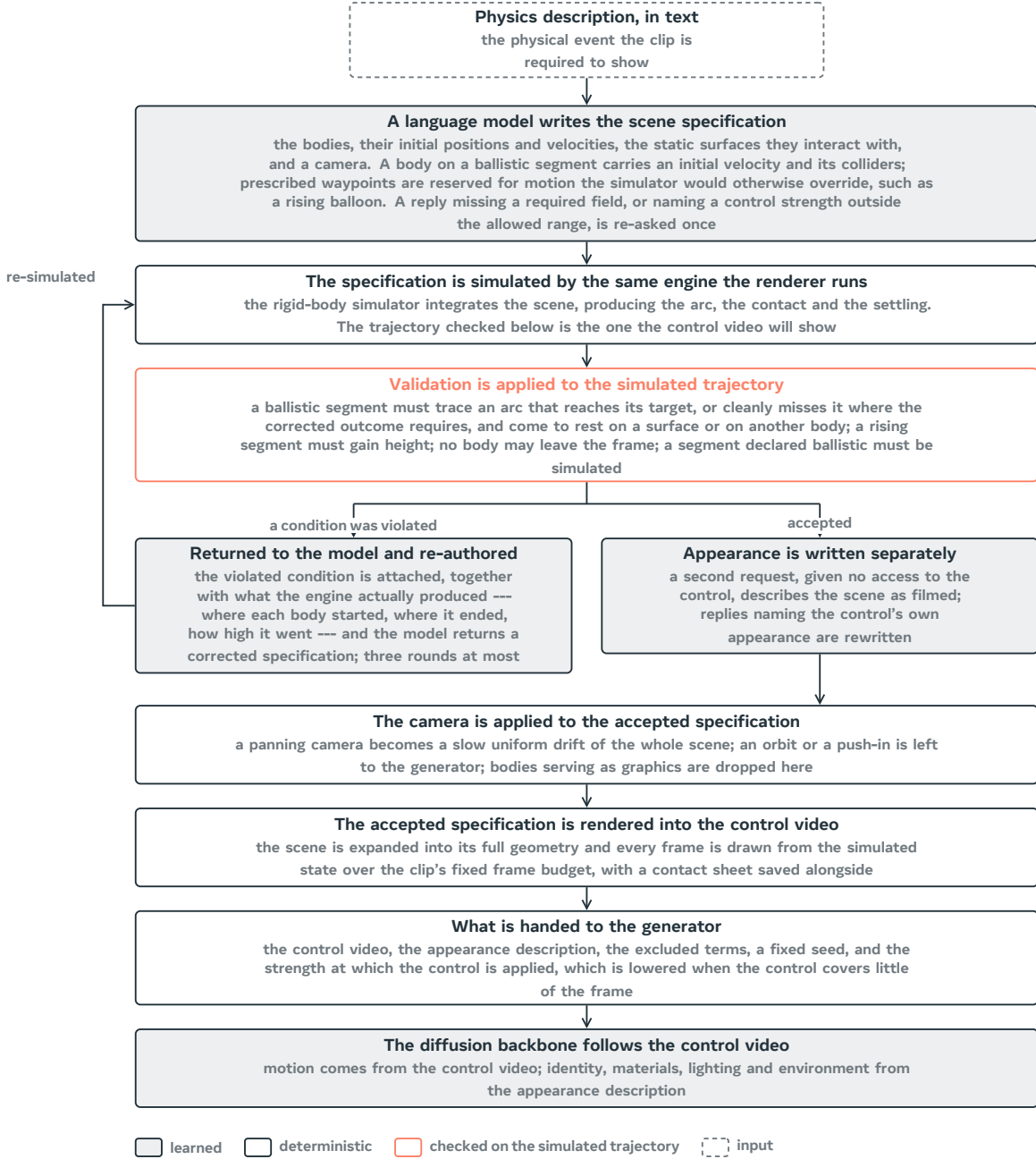


Figure 3 Authoring a control video from a physics description. A language model writes the scene specification; the same engine that renders the control simulates it; validation is applied to the simulated trajectory, and a failure returns to the model with the violated condition and a summary of the motion produced. The loop runs for at most three rounds. Appearance is written in a separate request and reaches the backbone through the text prompt.

model with a native control branch, run through the WanVACEPipeline interface. Conditioning synthesis on an explicit geometric signal is the mechanism introduced for images by ControlNet [79] and extended to motion and structure in video [29, 67]. The control used here originates in a physics solver, so the structure the generator follows satisfies the equations of motion by construction and carries the solver’s numerical error. The conditioning strength is distinct from the classifier-free guidance scale, which governs adherence to the text prompt and is held at the value in table 1 throughout. The control video is a grayscale clip in which brightness encodes distance from the camera, rendered from the simulation by the procedure of section 2.3. The prompt is a natural-language description of the scene’s subject, material and setting (“a bright red rubber

Setting	Value
Backbone	Wan 2.2-VACE-Fun-14B (two-expert MoE), diffusers Wan-VACEPipeline, bf16
Memory	experts CPU-offloaded (two-expert weights exceed an 80 GB accelerator)
Control signal	MuJoCo z-buffer depth (globally normalized); object silhouette for flat scenes
Conditioning	control video + all-white mask + text prompt; no reference image
Resolution	848×480
Frames / rate	81 frames at 16 fps
Sampler / steps	UniPC, 50 steps
Guidance scale	5.0
Conditioning scale s	1.0 deployed; $\{0.5, 0.8, 1.0\}$ examined separately

Table 1 Deployed generation configuration, from the generation scripts.

ball bouncing on a polished concrete studio floor”), paired with a single negative prompt shared by every clip whose terms are cartoon, CGI, render, grayscale and flat shading.

The mask decides which parts of the frame are held fixed and which are synthesised. Dark regions mark content to be preserved as given; bright regions mark content to be generated under the guidance of the control. Inside the backbone the mask splits the control video into a preserved part and a generated part, which are encoded separately and concatenated into the control latent stream. We pass an *all-white* mask, so the preserved part is empty and every pixel of every frame is synthesised. The depth field therefore contributes no pixel values to the output and acts only as guidance.

Geometry enters the network as a latent stream added into the denoising stream at each of the backbone’s control layers. Sampling proceeds over a decreasing sequence of noise levels $\sigma_1 > \sigma_2 > \dots > \sigma_N$, with $\sigma_N \rightarrow 0$, where σ_i is the standard deviation of the Gaussian noise still present in the latent at step i under the flow-matching parameterization of the backbone’s scheduler. Writing $h_i^{(\ell)}$ for the hidden state at layer ℓ and step i , z^{ctrl} for the encoded control video and $W^{(\ell)}$ for the control branch’s projection,

$$h_i^{(\ell)} \leftarrow h_i^{(\ell)} + s \lambda_i W^{(\ell)} z^{\text{ctrl}}, \quad \lambda_i = \begin{cases} 1, & \sigma_i > \sigma_{\text{rel}}, \\ 0, & \sigma_i \leq \sigma_{\text{rel}}. \end{cases} \quad (1)$$

The conditioning strength s is one scalar applied uniformly to every control layer. Lowering s weakens how tightly the generated motion is bound to the simulated motion without altering the appearance request. The gate λ_i withdraws the control once the latent is cleaner than the *release level* σ_{rel} , leaving the remaining steps unconditioned. The deployed recipe sets $\sigma_{\text{rel}} = 0$, so the control acts throughout; [section 2.5](#) varies it.

Geometry is supplied to the control branch as a separate conditioning input, under which the generated appearance is determined by the prompt rather than by the source render. One further setting was fixed empirically: the optional appearance reference is left unset, since supplying it degraded single-object trajectories under inspection.

[Table 1](#) lists the resulting configuration. Because the two expert stacks exceed the memory of a single 80 GB accelerator, the experts are CPU-offloaded during generation, and sampling is deterministic given the seed, which is recorded per clip. Generation runs as a staged GPU job. Control videos are rendered first. A 1.3B checkpoint then generates each scene once, and scenes whose output is degenerate are not queued for the 14B model. The 14B checkpoint generates the final clips, distributed across GPUs when more than one is available, and the clips are streamed to storage.

2.3 Rendering the simulation into a control signal

A control branch accepts a video rather than a scene description, so the simulation is rendered into one. The target is a grayscale video whose brightness encodes distance from the camera, and each decision below brings that encoding closer to the distribution of depth maps on which the control branch was trained.

One simulation, several registered views. A scene is described as data—a list of moving bodies with initial positions and velocities, a list of static surfaces, and a camera placed by a look-at rule—and the same engine turns any such description into a simulation. Dynamics are integrated in MuJoCo [64] under gravity at 9.81 m/s^2 with a one-millisecond solver step, a fourth-order integrator, and an elliptic friction cone. Output frames are produced on a coarser clock: the requested scene duration is divided into the frame budget, and the solver is advanced by that many sub-steps between frames. At every output frame, and from the *same* solver state, three renderers are queried in turn—a shaded colour view, the depth buffer, and the object-segmentation buffer. Because they read one state, the resulting videos are registered pixel for pixel, so a depth control and a silhouette control for the same scene describe the same geometry and can be substituted for one another without re-simulating. The control video is the simulator’s own depth buffer, so the geometry the generator follows is subject to the solver’s numerical error.

From depth buffer to control video. The depth buffer holds distance from the camera. An inverted, clipped affine map turns it into a control video in which near surfaces are bright and far surfaces dark, matching the polarity of the monocular depth maps the control branch was trained on. The endpoints are robust percentiles of the scene’s depth values, the first and the ninety-seventh, taken after discarding the far-clip constant the renderer writes for background pixels; without that exclusion a single distant constant compresses the whole range into a few grey levels. Formally, for depth frame D_t over pixel domain Ω and $t \in \{0, \dots, T-1\}$, let

$$\begin{aligned} \mathcal{D} &= \{D_t(u) : t \in \{0, \dots, T-1\}, u \in \Omega, D_t(u) \neq d_{\text{far}}\}, \\ d_- &= Q_{0.01}(\mathcal{D}), \quad d_+ = Q_{0.97}(\mathcal{D}), \\ C_t(u) &= 1 - \text{clip}\left(\frac{D_t(u) - d_-}{d_+ - d_-}, 0, 1\right). \end{aligned} \tag{2}$$

Here d_{far} is the renderer’s background constant and C_t is the grayscale control frame before boundary smoothing. If $d_+ = d_-$ the depth map is degenerate, and the silhouette control described below is used instead.

The mapping is global over the clip. The percentiles d_- and d_+ in [equation \(2\)](#) are computed over all T frames at once, making the assigned grey level a function of distance alone: a given distance maps to the same grey level in every frame, and a change in brightness indicates a change in distance. Per-frame normalization, the convention monocular estimators follow because they emit one relative map per image, breaks that correspondence—a body holding its distance while other content enters the view shifts brightness, which the control branch cannot distinguish from motion toward the camera. A per-frame variant is rendered alongside the global one for comparison on the same simulation.

Boundary treatment. A silhouette boundary in a simulator’s depth buffer is a one-pixel step, where the estimated depth maps the control branch was trained on have soft boundaries. Each grayscale frame therefore receives a small symmetric Gaussian blur before encoding. Symmetry leaves the half-intensity contour of an edge in place, so the smoothing alters the boundary’s appearance without displacing the geometry it encodes.

Silhouette controls for flat scenes. The segmentation buffer labels each pixel with the geometry that produced it. Painting the labels of the simulation’s moving bodies white on black yields a silhouette control: their outlines over time, carrying no distance information. Depth is the default, since it supplies both outline and distance ordering, but it loses geometric contrast on flat layouts, where every object rests on one near-horizontal surface at nearly the same camera distance. Measured on the rendered controls, the depth separation between a moving body and the surface around it is roughly twenty grey levels for a ball bouncing on a floor and a hundred for a ball arcing through open air, against fourteen for a rack of balls on a tabletop. Those layouts

Control applied for	Spread across clauses	Spread across seeds
no steps	0.875	0.252
the first 10 of 50 steps	0.060	0.427
all 50 steps	0.001	one seed

Table 2 Sideways travel varies with the prompt’s motion clause when the geometric control is absent, and with the seed when it is present. Each entry is a range (largest minus smallest) of travel in frame widths: the first column holds the seed fixed and varies the clause, the second holds the clause fixed and varies the seed. The control video is identical throughout. As the control acts over more of sampling the clause-attributable range falls to 0.001, while the seed-attributable range does not fall with it.

use the silhouette instead, which restores maximum figure-ground contrast at the cost of distance ordering between objects; the generator recovers that from the prompt and from scene priors.

A timing property of the renders. The frame budget and playback rate are fixed—eighty-one frames written at sixteen frames per second—while the simulated interval differs per scene. The clip therefore plays the simulated motion back stretched in time, by factors between roughly $1.8\times$ and $2.3\times$ for the scenes used here. The trajectory shapes and the contact events are the simulation’s; their absolute rate in the finished clip is not, and a downstream measurement of speed or duration must be read against the render clock rather than the simulator’s.

2.4 How much of the motion the control determines

The control need not act over the whole of sampling. Applying it and then withdrawing it at a chosen step gives two measurements: what the control determines while it is present, which is read by varying the prompt against it, and how long it must act, which is read by moving the withdrawal point. The first is measured below on the deployed configuration; the second requires a finer sweep and is measured on a second backbone (section 2.5).

What the control determines while present. What each input determines can be measured by holding one fixed and varying the other. The geometric control is the same projectile depth video throughout; the prompt varies over three wordings that share their subject, scene, camera, material and style and differ only in the motion clause. One states a strong horizontal traverse, one is neutral, and one is adversarial with respect to the control, requiring the object to drop nearly straight down and bounce in place with almost no sideways motion. Each prompt is generated under three conditioning regimes: no control at any step, control over the first 10 of 50 steps, and control throughout, at three seeds for the partial regime, two for the unconditioned one and one for the full regime. The measured quantity is the horizontal distance the object travels as a fraction of frame width, recovered by tracking it in the finished clip. The metric and the decision rules were registered before generation.

The result is a crossover in what the outcome varies with (table 2 and figure 4). Without a control the prompt determines the motion, travelling 0.05 of the frame under the adversarial clause and 0.97 under the horizontal one, a range of 0.875 between clauses against 0.252 between seeds. The unconditioned clips also differ from the conditioned ones in framing: the model places the camera close to the ground with no horizon and renders the ball at a median tracked area of 8,846 pixels, against 211 under the control. Travel measured as a fraction of frame width is therefore not a like-for-like comparison across the first and third columns, and the within-column comparisons carry the argument. The ordering of the three clauses holds within each seed, while the magnitude does not reproduce for the neutral clause, which spans 0.21 and 0.86 over its two seeds. With the control applied throughout, the three prompts produce travels of 0.872, 0.871 and 0.871: the clause-attributable range has fallen to 0.001, at one seed per clause. In between, with the control applied over the first fifth of sampling and then withdrawn, the two sources exchange places: the clause-attributable range is 0.060 and the seed-attributable range 0.427, seven times larger.

The trajectories locate where that shortfall arises (figure 4). A clip conditioned over the first ten steps does

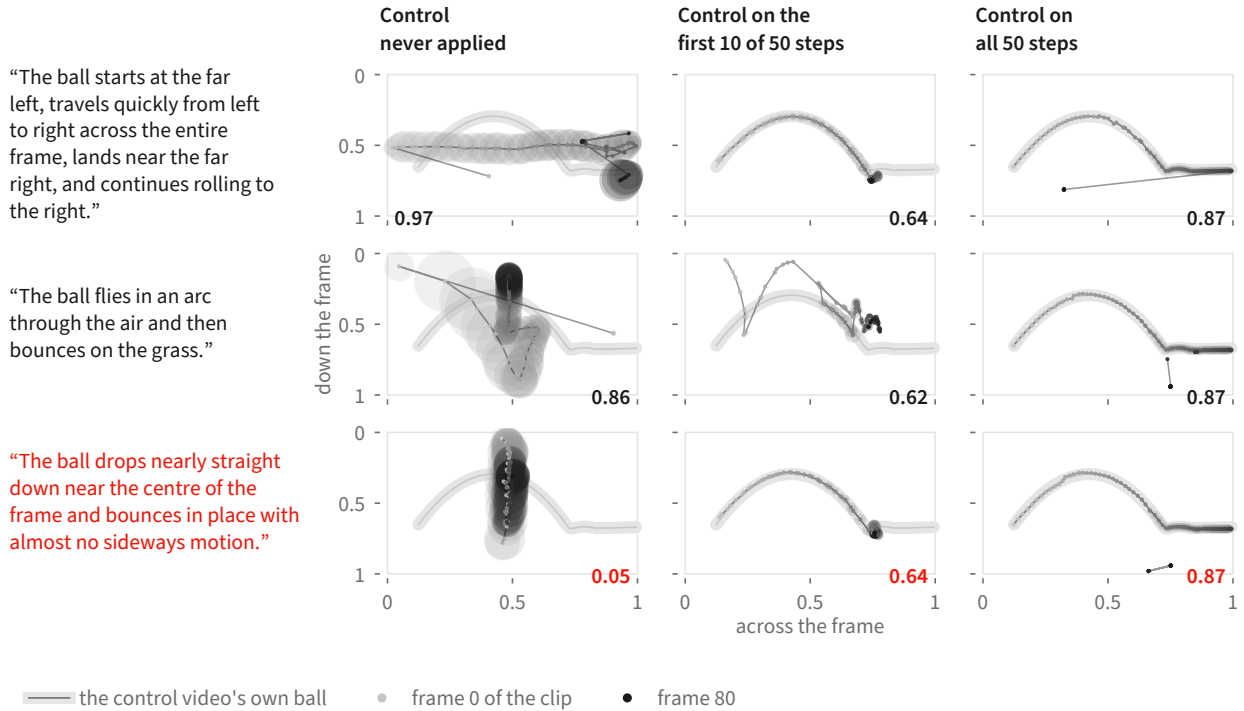


Figure 4 The tracked path of the ball in nine clips at one seed, against the same depth control in every one. Each box is a single 848×480 frame with x running left to right and y downward; one dot is one frame, drawn at the area of the region the tracker located and shaded from the first frame to the last. The pale band is the ball in the control video, in frame for 63 of its 81 frames. The number in each box is that clip’s sideways travel. The three prompts differ only in the closing sentence set beside each row, and the row in red asks for the motion the control does not perform. Without the control the path follows the clause, and the camera framing changes with it. With the control throughout, all three clauses reproduce the control’s arc. With the control over the first ten steps the path holds the arc and departs at the landing.

not leave the control’s path when the control is withdrawn: it tracks the arc through the flight and departs at the landing, stopping near 0.77 of the frame while the control’s ball rolls on to 0.99 and out of frame. The lost travel is the roll after contact rather than a different flight, which is the same phase that [section 2.5](#) finds most sensitive to an early withdrawal. Of the three clauses at this seed, two follow the control from the first frames and the neutral one does not, remaining a mean 0.243 of the frame from the control’s path over the first ten frames. The trajectory established while the control is present persists through the remaining steps, during which the text is the only conditioning.

2.5 How long the control must act

Moving the withdrawal point locates the noise level at which the simulated trajectory becomes fixed. These measurements were made on a second Wan2.2-derived backbone, run at 40 UniPC steps with a flow shift of 5.0. It has no trained control branch: its only conditioning path encodes a rendered source video into context tokens concatenated with the text and image tokens. That makes the withdrawal directly available — the source tokens are removed from the attended sequence at a chosen step, a different operation from the gate of [equation \(1\)](#) — and it is the configuration on which the schedule of [equation \(4\)](#) and the expert switch below are defined. The source render is encoded into video tokens that are concatenated with the image and text tokens the sampler attends over, each stream carrying a scale ω . Withdrawal removes the video tokens from the sequence outright, rather than scaling an additive residual to zero: writing $m_i \in \{0, 1\}$ for their presence

at step i of S ,

$$m_i = \mathbb{K}[i < i^*], \quad i^* = \text{clip}(\text{round}(fS), 0, S - 1), \quad (3)$$

with f the withdrawal fraction. Steps before i^* attend over the source tokens at the scale ω the deployed recipe uses; from i^* onward the sequence is identical to that of an unconditioned generation.

Because i^* is an index, the noise level it corresponds to depends on the schedule. The backbone samples on the flow-matching schedule of its UniPC solver, so for S steps at shift κ over N training timesteps,

$$\tilde{s}_i = \frac{(S - i)(1 - 1/N)}{S}, \quad \sigma_i = \frac{\kappa \tilde{s}_i}{1 + (\kappa - 1) \tilde{s}_i}, \quad \sigma_{\text{rel}} = \sigma_{i^*}. \quad (4)$$

Two runs that withdraw at the same step index therefore correspond to different noise levels whenever S or κ differs, .

The solver is multistep and forms each update from derivatives cached at previous steps, so after a withdrawal those cached values still carry the influence of the removed tokens. One of the two variants below clears that history at i^* and the other leaves it in place, and the two are otherwise identical.

The control is a shaded-colour render rather than a depth render, and the scenes consist of a single moving rigid body at one resolution and frame count, chosen because their trajectories can be checked against a closed-form path. One of the three yields only five measurable frames after ground contact, so the windowed comparisons rest on two, and each schedule setting carries one seed.

Withdrawing the conditioning partway leaves the remaining steps unconditioned. Noise decreases as sampling proceeds, so a *high* σ_{rel} is an *early* withdrawal and a low one a late withdrawal; that convention is used throughout. A clip that still follows the simulated trajectory after an early withdrawal shows the trajectory was determined before the control was removed, so the earliest withdrawal that preserves agreement marks where the trajectory becomes fixed.

Three runs are reported, each answering a different question. The first fixes one withdrawal point and repeats it over 16 seeds, giving the seed-to-seed variability. The second sweeps six withdrawal points on one seed, giving the shape of the transition. The third re-scores a denser sweep with an independent tracker, testing whether the effect depends on how the object is measured. The quantities they report—retention, its two windows, and the two failure counts—are defined next.

Method. Each run fixes one σ_{rel} and is paired against a run at $\sigma_{\text{rel}} = 0$ sharing its simulation, control video, prompt, sampler and seed. Release points are indexed by σ_{rel} .

Both the control video and the generated clip are tracked frame by frame, giving the object’s path in each. The default tracker locates the moving body by its colour against the background and reports the centroid of the largest matching region; the third run below substitutes a segmentation-based tracker, which takes the centroid of the object’s predicted mask instead.

Write $p_t^{\text{gen}} = (x_t^{\text{gen}}, y_t^{\text{gen}})$ and p_t^{ctrl} for the tracked positions at frame $t \in \{0, \dots, T - 1\}$, undefined on frames where the tracker locates nothing, and let $V = \{t : p_t^{\text{gen}} \text{ and } p_t^{\text{ctrl}} \text{ both defined}\}$. Retention over a window W is the Pearson correlation of the two coordinate series, taken separately per axis over the frames of W at which both are located,

$$R_x(W) = \rho(\{x_t^{\text{gen}}\}_{t \in W \cap V}, \{x_t^{\text{ctrl}}\}_{t \in W \cap V}), \quad R_y(W) = \rho(\{y_t^{\text{gen}}\}_{t \in W \cap V}, \{y_t^{\text{ctrl}}\}_{t \in W \cap V}), \quad (5)$$

reported only where $|W \cap V|$ reaches a minimum count. The intersection with V is what makes a frame the tracker misses in either video drop out of both series rather than either one alone.

The two windows are separated at the first frame of ground contact, read off the control’s own vertical path rather than set per scene. With y^{ctrl} measured downward and $t_{\text{peak}} = \arg \max_{t \in V} y_t^{\text{ctrl}}$ the object’s highest point,

$$t_c = \min \left\{ t \geq t_{\text{peak}} : y_t^{\text{ctrl}} \leq y_{\min} + \frac{1}{10}(y_{\max} - y_{\min}) \right\}, \quad (6)$$

where $y_{\min}$ and $y_{\max}$ are taken over the located frames of the control. The *airborne* window is $W_{\text{air}} = \{t < t_c\}$ and the *after-contact* window $W_{\text{post}} = \{t \geq t_c\}$. A value of R near 1 means the generated object moved as

At one release point (noise level 0.95), 16 seeds per bar

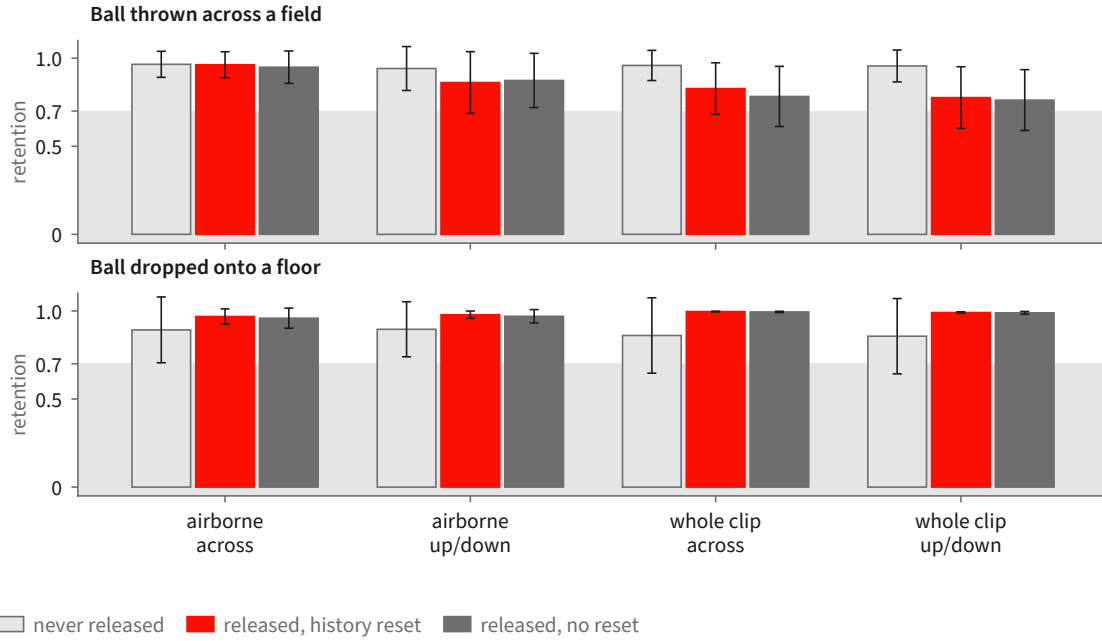


Figure 5 The seeded measurement at the single release point $\sigma = 0.952$, 16 seeds per bar, error bars showing 95% confidence intervals of the mean. Retention is the correlation between the generated object’s tracked path and the simulation’s, as defined in the text. The three bars in each group are the never-released baseline and the two withdrawal variants, which differ only in whether the sampler’s cached derivatives are cleared at the withdrawal. Grey band marks the region below the 0.7 threshold fixed before scoring.

the simulation did. Two windows are reported, since the windowed metrics localise a disagreement that a single whole-clip value does not.

Two failure modes are counted alongside retention. A *whole-clip disagreement* is a clip whose horizontal agreement over the full clip falls below 0.7. A *late-window disagreement* is a clip meeting two conditions at once, whole-clip horizontal agreement below 0.7 and airborne horizontal agreement above 0.9. Since the airborne window agrees while the clip as a whole does not, the disagreement is located in the remaining frames. Both thresholds were fixed before scoring.

Airborne and after-contact agreement under an early release. Two scenes were repeated over 16 seeds each at one withdrawal point, $\sigma = 0.952$, in two variants—with and without clearing the multi-step sampler’s cached previous-step derivatives at the moment of withdrawal—against the never-released baseline: 96 clips in total, 16 per cell (figure 5). Releasing the control at $\sigma = 0.952$ leaves the airborne flight of the thrown ball intact, with per-seed horizontal agreement dropping by 0.003 relative to never releasing (95% confidence interval 0.001 to 0.004, $n = 16$ paired seeds). Over the whole clip the same withdrawal reduces horizontal agreement by 0.13 (95% CI -0.02 to 0.28) and 0.18 vertically (95% CI 0.07 to 0.29). Agreement therefore differs sharply between the two windows: the airborne arc is unaffected and the disturbance is confined to the motion after the ball reaches the ground. Scoring only the frames after the simulated contact, the generated ball’s horizontal motion runs *opposite* to the simulation’s on 9 of 16 seeds at $\sigma = 0.952$ and on 0 of 16 when the control is never released—the simulation moves the ball forward and the unconditioned generator moves it backward. Whole-clip disagreements number 3 of 16 seeds against 1 of 16, and late-window disagreements 2 of 16 against 0 of 16.

Release-point sweep across three scenes. Sweeping the withdrawal point across six settings on a single seed for three scenes estimates retention as a function of withdrawal noise level (figure 6). For the thrown ball,

Retention against the release point

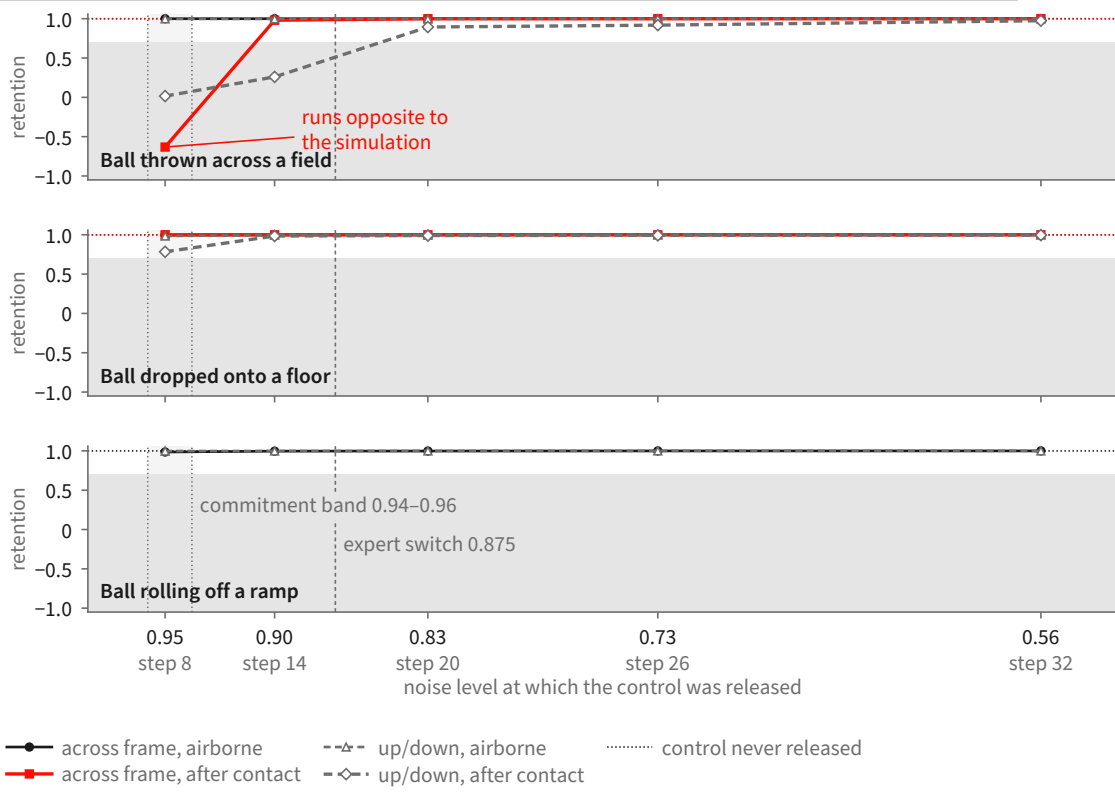


Figure 6 The release-point sweep: one clip per point, three scenes, scored separately over the airborne window and the window after the simulated object first reaches the ground. The horizontal axis is the noise level at which the control was switched off, high noise on the left; the step index of each run is printed underneath. The thrown ball’s after-contact agreement falls below the threshold in both directions at the earliest withdrawal, and the vertical component is the later of the two to recover. Guides mark the commitment band and the backbone’s expert switch; the grey band marks the region below the 0.7 threshold fixed before scoring.

releasing at $\sigma = 0.952$ gives an after-contact horizontal agreement of -0.63 , indicating motion opposite to the simulation’s, with the airborne window still at 1.00; releasing at $\sigma = 0.903$ or later restores the after-contact window to 0.98 or above, and it stays there through the latest release point tested, $\sigma = 0.555$. The same sweep on a ball dropped onto a floor stays above the threshold at every setting: after-contact horizontal agreement is 1.00 at every release point including the earliest, and the vertical agreement, the informative direction for a straight-down drop, is lowest at the earliest release, 0.78 at $\sigma = 0.952$, rising to 0.99 or above from $\sigma = 0.903$ onward. A ball rolling off a ramp remains at 0.99 or above at every withdrawal point in both directions; a locatable object is present in about a third of its frames, and each of its correlations is computed over that subset. Across all three scenes, the vertical component retains agreement at earlier withdrawal points than the horizontal component.

Replicated withdrawal points under a different tracker. A third run covers fewer withdrawal points than the sweep above but replicates each of them. It takes three withdrawal points on the thrown ball at three seeds and two classifier-free guidance settings, 18 clips in all, and scores them with the segmentation-based tracker rather than the colour tracker. It reproduces the direction of the effect: airborne horizontal agreement is 1.00 (mean over the 6 clips at $\sigma = 0.952$, 95% CI 0.998 to 1.000), while after-contact horizontal agreement at the same release point averages 0.69 with a confidence interval spanning zero (-0.04 to 1.43 , $n = 6$) and is negative on one clip; at $\sigma = 0.903$ and $\sigma = 0.833$ the after-contact window recovers to 0.99 and 1.00. At the latest withdrawal point the tracker locates no object on 3 of 6 clips; manual visual inspection of those clips shows a ball of low contrast relative to the grass background. Clips yielding no location are excluded, so that

cell’s mean and interval are computed over 3 clips and are correspondingly wide.

Location of the commitment point. Across the runs the trajectory stops responding to removal of the control in a narrow band of noise, in the region of $\sigma \approx 0.94$ to 0.96 . Under [equation \(4\)](#) at $S = 40$ and $\kappa = 5.0$ that band lies in the high-noise portion of sampling, consistent with the critical-window account [5, 34, 56]. It also lies above the backbone’s own architectural boundary: the model runs two expert networks and switches from the high-noise to the low-noise one when the timestep falls below $0.875 N$ [14], so the motion is fixed before the handoff occurs.

Of the six clips scored at the schedule settings, two diverge from the reference and one is borderline. Three yield no usable track, so the band’s width is not estimated. Within it the flight arc and the ground contact are not separated: both recover at the same band as the withdrawal is moved later.

Under [equation \(4\)](#) a change in S or κ moves the step index at which the band falls while leaving the noise level unchanged, which is the reason for the coordinate adopted here. Prior work on partial conditioning organises the process by noise level for the same reason [10, 43].

Which component recovers last. Separating the two windows shows that the components of the motion do not become fixed together. On the thrown ball, withdrawing at $\sigma = 0.9522$ drops after-contact agreement below the threshold in both coordinates, horizontal to -0.632 and vertical to $+0.016$. Moving the withdrawal one point later, to $\sigma = 0.9025$, restores the horizontal component to $+0.981$ while the vertical is still at $+0.260$; the vertical does not reach $+0.894$ until $\sigma = 0.8331$ and $+0.975$ until $\sigma = 0.5552$. The ball dropped onto a floor shows the same ordering at the single withdrawal point where its agreement falls, horizontal $+1.000$ against vertical $+0.784$ at $\sigma = 0.9522$. A withdrawal schedule set by the horizontal component alone would therefore be too early for the motion that remains.

Over the whole clip the thrown ball’s vertical agreement is $+0.986$ against a horizontal $+0.421$, the reverse ordering. That figure is produced by the large up-and-down excursion of the flight, which dominates the correlation and leaves it insensitive to the rebound that follows.

The contact as the limiting phase. The failure an early withdrawal produces is specific: the flight is reproduced and the behaviour after ground contact is not, with the ball reversing direction on 9 of 16 seeds at $\sigma = 0.952$. Contacts and collisions are a documented failure mode of video generators [3], which locates the mechanism in the strength of the model’s prior over contact. Where that prior is weak the control must remain present to override the default it supplies: the baseline with the control applied throughout reproduces the contact correctly on every seed, and the reversal appears only when the guidance is withdrawn early.

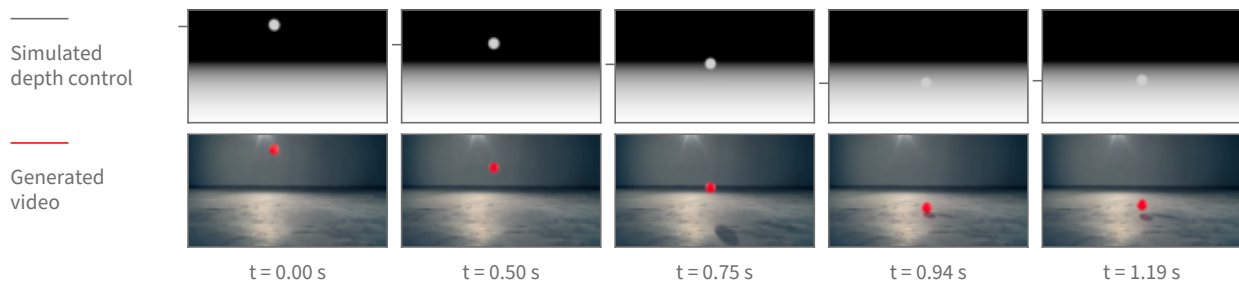
Sampling coverage and the solver reset. Clearing the sampler’s cached derivatives changes the result by at most 0.04 horizontally over the whole clip (95% CI -0.04 to 0.13 , $n = 16$), the largest per-seed difference on any window or direction measured.

2.6 Coverage of the scene library

The same pipeline applies to scenes of higher body count than the commitment analysis covers. It has produced 1,314 scenes across 66 kinds, where a *kind* is a scene template and a *scene* is one parameter setting of a kind, simulated once and generated once. Before any photorealistic render, 652 of the simulations were inspected and 591 of those 652 judged to show the intended physical behaviour clearly enough to generate from. The inspection covered simulator output; generated clips were not scored in it.

A single authoring, simulation and conditioning path covers body counts from one to several dozen and several classes of physics. [Figure 8](#) shows twelve of them as generated frames: cloth draping and crumpling, a flag in wind, a swinging rope and a hanging chain, grains pouring into a heap and draining through a funnel, a ball dropped into a pit of balls, a loaded boat, a cascade down a slope, toppling dominoes and a chaotic two-arm linkage. The body counts are the scenes’ own parameters, reaching 34 small bodies for the pouring heap, 30 for the funnel, 20 rigid slabs for the dominoes, 9 in contact for the ball pit and 6 on the ramp for the cascade. [Figure 9](#) shows three of them as depth control and generated clip at matched timestamps; the three are those

Bouncing ball



Ball thrown across a field

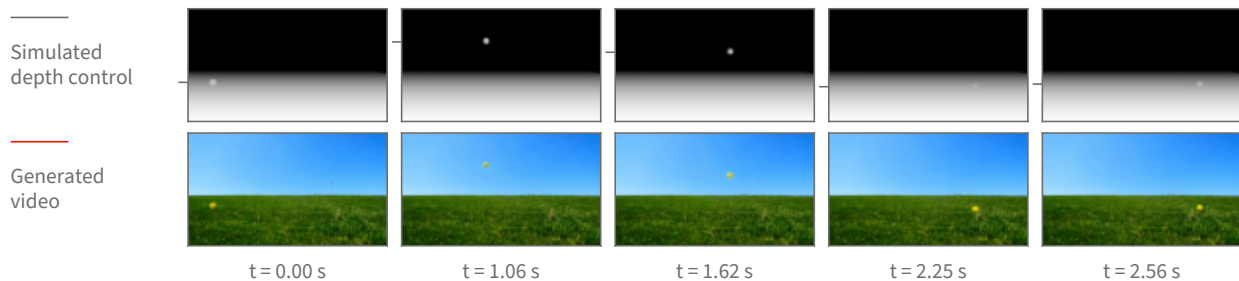


Figure 7 The same mechanism across two scenes with very different requested appearance—an indoor studio and an outdoor field—each shown as its depth control above and the resulting generated clip below. Each scene has its own timestamps, chosen from the turning points of that scene’s tracked vertical path, so each column falls within a moving interval. The appearance changes completely with the prompt while the motion stays the simulation’s: for the thrown ball the generated and control paths agree to a correlation of 1.0000 horizontally and 0.9998 vertically, with a median disagreement of 0.7 pixels across and 1.1 pixels down, over the 63 frames in which the ball is measurable in both before it leaves the frame.

whose depth control stays distinguishable from its background throughout, since an object at rest on the floor takes the floor’s grey and leaves the control blank.

Fidelity does not extend uniformly across that range. On a small set of scenes assessed by manual visual inspection together with the colour tracker of [section 2.5](#), single-object scenes reproduce the simulated trajectory, with the generated object following the control throughout ([figure 7](#)); a multi-object interaction such as a billiards break does not, and neither the deployed backbone nor the alternates tried against it produced a collision in which every body moved as the simulation specified. The commitment point for multi-body motion is likewise unmeasured, since the colour tracker cannot follow a scatter as a single path.

Every scene in the library is produced by the same rigid-body engine, and the deformable and granular kinds are built from that engine’s particle-and-link primitives: cloth is a sheet of particles joined by springs, a rope or chain is a short row of linked segments, sand and marbles are many small solid bodies in contact, and floating is a buoyancy force on a solid body in a still fluid volume. The library therefore covers these primitives and not fluid, smoke or finite-element soft-body solves.

Beyond a single ball

12 of 66 scene kinds

One frame of the generated video per scene, at the point half the clip's motion has happened.



Cloth draped over a cylinder
particle sheet



Cloth falling, crumpling
particle sheet



Flag rippling in wind
pinned particle sheet



Rope swinging, whipping
linked segments



Chain hanging, swinging
linked segments



Grains pouring in a heap
34 small bodies



Grains draining a funnel
30 small bodies



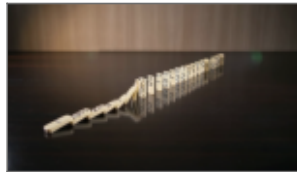
Ball into a pit of balls
9 bodies in contact



Loaded boat stays afloat
buoyancy force



Cluster down a slope
6 bodies on a ramp



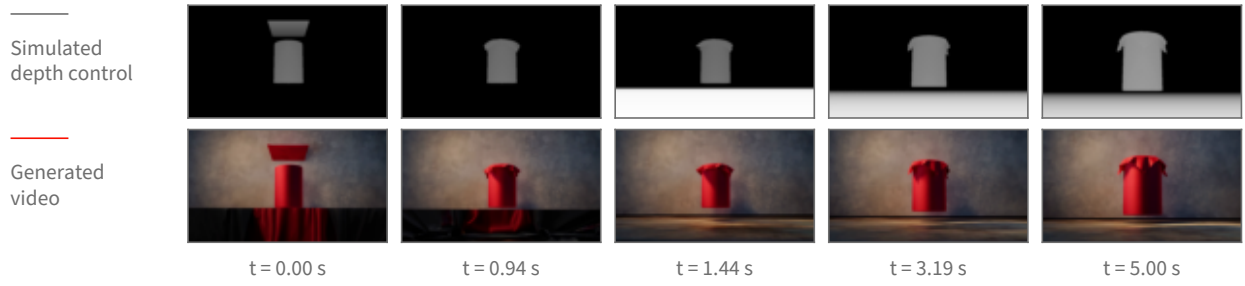
Dominoes toppling in turn
20 rigid slabs



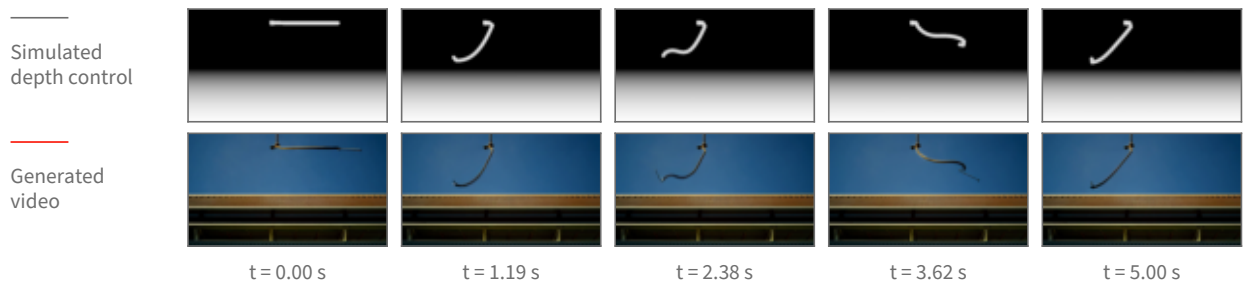
Linkage swinging chaotically
two jointed arms

Figure 8 Twelve of the 66 scene kinds, one generated frame each. Each tile is selected by rule: the frame-to-frame change is accumulated over the whole clip and the tile is the frame by which half of that total has occurred, which falls during the scene's principal motion rather than at its start. The second caption line names the physics primitive, and every body count in it is the scene's own parameter. Each clip's simulation passed the inspection described in the text, and the tiles were additionally inspected manually.

Cloth falling and draping over a cylinder



Rope swinging from a fixed anchor



Grains draining through a funnel

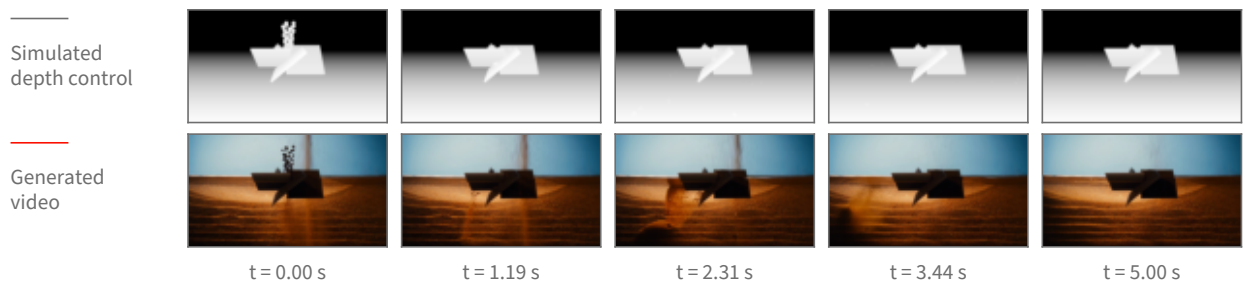


Figure 9 Three deformable and granular scenes, each with the simulated depth control above and the generated clip below, at matched timestamps. Columns are placed at equal steps of the clip’s accumulated change, so none falls in a temporally static interval. Pairs are shown only for scenes whose depth control remains distinguishable from its background in every column; an object that settles onto the floor takes the same grey as the floor and leaves the control blank.

3 A Human-Annotated Video-Flaw Benchmark

The benchmark is a corpus of 1,500 AI-generated clips in which a person watched each clip against its prompt and wrote down every place the video failed to deliver what the text asked for. Every clip carries at least one such record. The generating model is not recorded in the data, and no clip is attributed to a specific system. The human-written fields are the ground truth throughout this report.

3.1 What a flaw record contains

A flaw is a localized record with six fields. Four are always present: the **violated fragment**, the words of the prompt the video fails, quoted verbatim; a free-text **rationale**; a **severity**; and the annotator’s **confidence**. Two are present when the annotator could supply them: a **time span**, the first and last frame in which the failure is visible, and a **box track**, boxes at the start, middle and end of that span, one per object involved, in normalized coordinates. [Table 3](#) gives each field’s measured coverage across the evaluation core, and [figure 10](#) shows one record in full. Each clip additionally carries an overall severity and a short list of the objects, actions and relations its prompt calls for.

Two coarse tags support grouping and reporting: a *modality* tag saying whether the complaint concerns what is seen, what is heard, or both, and a *type* tag naming the dominant kind of complaint from a fixed ten-item list. Neither is written by the annotator. Both are assigned by three language models classifying each flaw from its quoted fragment and rationale at temperature zero under a fixed seed, resolved by majority; a three-way disagreement resolves to “both” for modality and “other” for type, and a flaw receiving no usable vote falls back to a frozen keyword rule. They are labelled machine-assigned wherever they appear and are never treated as ground truth. Annotation is single-annotator: the corpus carries one annotation per flaw, so no inter-annotator agreement figure is available.

3.2 How the evaluation set is derived

Three recorded stages narrow 1,500 clips to the 149 every recall figure in this report is measured over. [Table 4](#) gives the counts and [figure 11](#) the same derivation as a sequence of steps.

Stage one, clip selection. A clip enters the pool if its video is retrievable and it shows signs of dynamics. The dynamics test is a fixed vocabulary of motion words—falling, bouncing, rolling, colliding, pouring, toppling—applied to the prompt and, separately, to each flaw’s rationale and quoted fragment; a clip is kept when its prompt contains one, or one of its flaws does, or it has a time-windowed flaw against an action-typed object. The test is a keyword rule and selects for prompts and complaints that *talk about* motion, which is a proxy for, not a guarantee of, a flaw whose adjudication needs physical reasoning. This stage keeps 606 of the 1,500 clips, carrying 1,171 flaws.

Stage two, flaw selection. Within those clips, a flaw is a usable detection target only if a critic seeing only pixels could in principle find it. Flaws tagged “seen” or “both” are kept and purely audible complaints dropped, which removes 64 flaws and leaves 1,107 as the full evaluation set.

Stage three, the frozen core. A core of 150 clips with 306 flaws is drawn for routine measurement. The draw is deterministic and its contract recorded: a fixed seed; strata crossing overall clip severity, flaw count per clip and clip duration; largest-remainder proportional allocation across strata; within each stratum, candidates ordered lexicographically then shuffled once by a generator whose call sequence is fixed; and a replacement rule drawing from the same stratum for any clip that cannot be downloaded or decoded. A continuity-focused stratum of 29 clips and 85 flaws sits inside the core and is exempt from replacement. Every clip is downloaded and decoded before the core is written, and the selection, strata and replacements are stored with the set.

The core is an *optimization set*: system and prompt decisions are made against it, so results on it measure fit to that core, and a generalization claim requires a set that has not influenced system design. A separate pool of 711 annotated clips is held disjoint from both frozen sets for any learning that consumes this data.

One clip of the 150 produced no measurement in the evaluated run and is absent from every figure in [section 5](#): a ten-second classroom scene carrying two flaws, both marked at the highest severity. Recall is therefore reported over 149 clips and the 304 flaw entries they carry, and never over the full 306. Those two flaws are

Field	What it holds	Coverage
violated fragment	the words of the prompt the video fails, quoted verbatim; median 8 words	306/306
rationale	free text saying what went wrong instead; median 25 words	306/306
severity	1–5; the distribution is centred at 3 and 4	306/306
confidence	1–5; 239 of 306 are marked 4 or 5	306/306
time span	first and last frame in which the failure is visible	237/306
box track	boxes at the start, middle and end of that span, one per object involved, in normalized coordinates	208/306

Table 3 The six fields of a flaw record and how often each is filled across the 150-clip core. The first four are always present; timing and boxes are not, and a flaw carrying neither is one whose complaint is about the clip as a whole or about something that never appears.

What one human flaw record contains

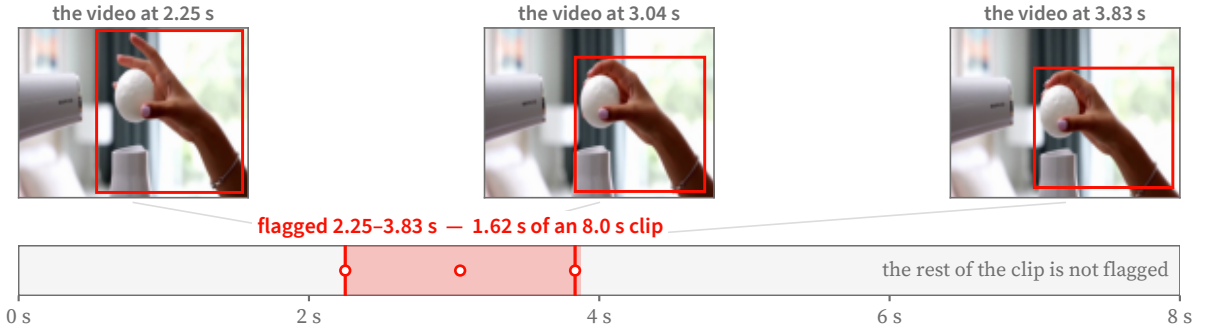
one of the 306 records in the evaluation core

THE PROMPT THE VIDEO WAS ASKED TO SATISFY

A ping pong ball balanced on a hair dryer's air stream, wobbling but never falling. A hand tries to grab it and misses. Audio: Dryer whirl, laughing.

the fragment the video fails to satisfy

WHERE THE FLAW OCCURS



HOW BAD IT IS, AND WHY

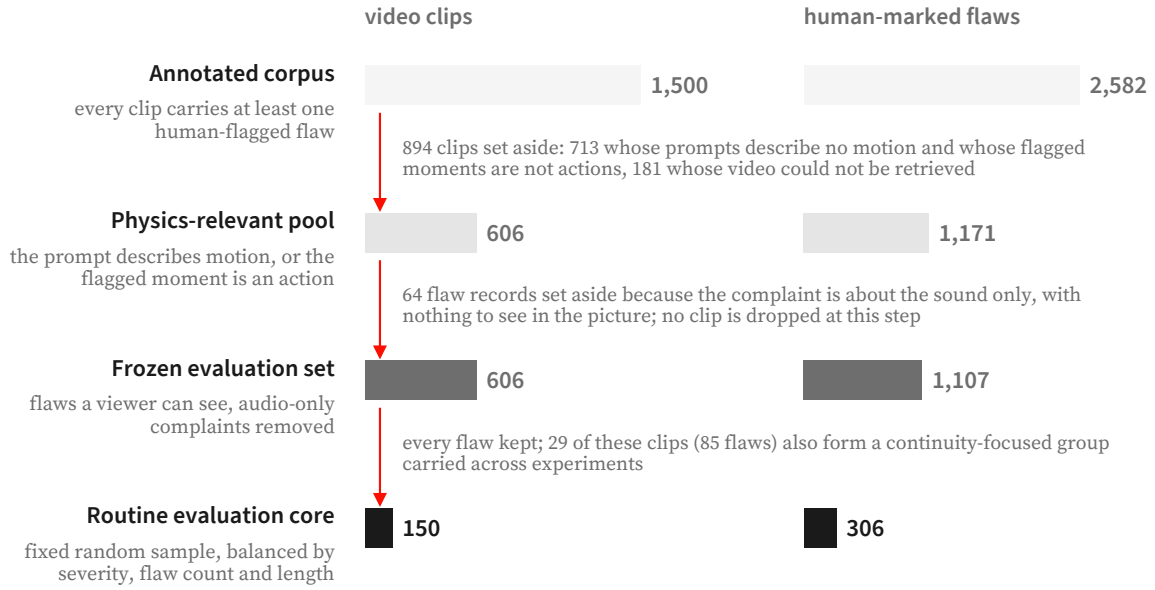
severity ●●●○○ 3 of 5 annotator confidence 5 of 5

the annotator's own reasoning, as written

"The prompt requires "a hand tries to grab it but fails", meaning the object should not be caught. However, it is caught in the actual video, which does not match the requested result."

Figure 10 One human flaw record, shown in full. The annotator quotes the prompt fragment the video fails to satisfy, bounds the interval in which the failure is visible, boxes the objects involved at the start, middle and end of that interval, and writes a severity, a confidence and a rationale in their own words. Frames and boxes are the annotator's; the flagged window is 1.62s of an 8s clip. Nothing about the critic's output appears here.

excluded rather than counted as missed, because nothing was measured on that clip either way, which makes the reported rate very slightly optimistic.



A further 711 annotated clips (1,130 flaws) are held back for training and share no clip with the frozen evaluation set or its routine core. Bar lengths are to scale within each column.

Figure 11 How the evaluation set is derived from the annotated corpus. Clips are removed only at the first step; the second step removes flaw *records* whose complaint is audible but not visible, leaving the clip count unchanged. The routine core is a fixed random sample balanced by severity, flaw count and clip length. A further 711 annotated clips are kept aside for training and share no clip with either frozen set.

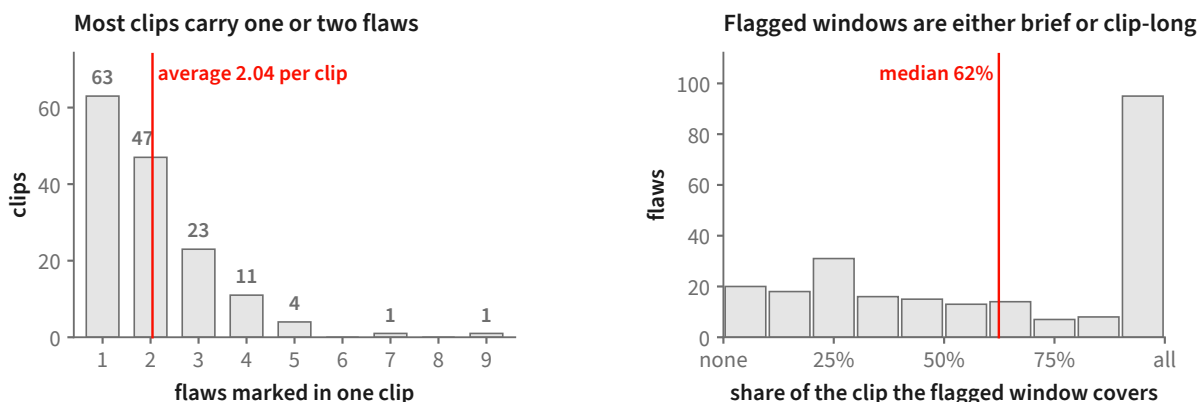
Set	Clips	Human flaws
Full corpus (all AI-generated, all annotated)	1,500	2,582
Physics-relevant pool	606	1,171
Full evaluation set (frozen)	606	1,107
Routine evaluation core (frozen)	150	306
continuity stratum within the core	29	85
Disjoint training pool (excludes the frozen sets)	711	—

Table 4 Benchmark statistics, from the dataset manifests. The full evaluation set drops a small number of audio-scoped flaws relative to the physics-relevant pool. The training pool excludes every clip in the frozen evaluation sets.

3.3 What is in the evaluation core

The core is 150 clips, median 8.0 s long (range 5.0–10.3 s), mostly at 24 fps (118 of 150; the rest at 16, 25, 30 or 32). The prompts are not one-line captions: a median prompt is 42 words and the longest 188, and many specify a shot list, a count, an ordering, an on-screen string and a sound in one paragraph. Sixty-six of the 150 clips carry flaws of more than one kind, so a single number per clip cannot say which of them a detector found.

Its 306 flaws, a median of two per clip, group by machine-assigned type as: actions that do not happen or happen wrongly, 139; wrong counts, 36; wrong or missing objects, 30; garbled on-screen text, 25; wrong spatial arrangement, 19; wrong ordering in time, 17; wrong attributes such as colour or material, 16; camera and style, 10; physical motion, 7; and a residual 7. The largest category is whether the requested action occurred at all, so this is not a physics benchmark in the narrow sense. Counting, text and spatial arrangement together account for 80 flaws, each a question a general-purpose model answers poorly and a dedicated measurement



150 clips of the routine evaluation core, 306 human-marked flaws. The right-hand panel uses the 237 flaws that carry a start and an end frame; the remaining 69 were marked without one. Of those 237, 57 fit inside a quarter of their clip and 95 run almost its whole length; the median flagged window lasts 5.00 s.

Figure 12 Two properties of the routine evaluation core that shape how a detector must behave. **Left:** most clips are flagged in only one or two places, so a detector that raises many complaints per clip will mostly be raising ones no annotator marked. **Right:** flagged windows are bimodal — many failures are confined to a small part of the clip while a comparable group runs almost its whole length — so both a brief localized check and a whole-clip check are needed.

answers well.

Of the 237 flaws carrying a time span, the flagged window covers a median of 62% of the clip, and the distribution is bimodal: 57 are confined to a quarter of the clip or less, and 95 run for at least nine-tenths of it (figure 12). A detector needs both a way to look closely at a moment and a way to judge a whole clip.

3.4 Scoring conventions

Every clip carries at least one flaw. With no clean clips, recall is directly measurable and precision is not: a detector that flags everything scores perfectly on recall alone, so results on this set are reported alongside the rate at which accusations land on a real flaw.

Two counts describe the same core, and they differ. Eleven clips record the same failure two or three times under different identifiers, seventeen duplicate entries in all, so the core’s 306 entries are 289 distinct flaws. Recall in section 5 is reported over entries rather than distinct spans, because the matching protocol there assigns each finding to at most one entry and a duplicate entry can be claimed only once; the 149 clips it scores carry 304 of the 306.

3.5 Relation to existing video benchmarks

The closest annotated resource is Physion-Eval [80], which pairs generated clips with real-world reference processes and provides expert reasoning traces containing temporally localized glitches, a physical-failure category, and an explanation. VideoPhy-2 [4] instead has three annotators rate semantic adherence and physical commonsense, and mark predefined physical rules as violated, followed, or undetermined. Per-dimension suites such as VBench [28] and T2V-CompBench [62] primarily support comparison among generators on fixed prompt collections.

Our corpus targets a complementary, prompt-grounded detector protocol. Each record attaches a failure to the exact words of the generation prompt and permits a time span and box track, so an allegation can be matched to a particular annotated failure rather than only correlated with a clip-level score. Unlike Physion-Eval, this protocol does not require a matched real-world reference video; unlike VideoPhy-2, it uses open-vocabulary flaw statements rather than a fixed candidate-rule list. These choices make claim-level

miss attribution possible in [section 5](#), but they do not improve annotation reliability: the present corpus is single-annotator and contains only flawed clips, whereas VideoPhy-2 provides redundant judgments and majority resolution.

4 PhyReAct: Auditable Physical Verification

4.1 PhyReAct as a physically grounded ReAct instance

PhyReAct inherits the control structure of the ReAct paradigm [77]: a text-only reasoner interleaves *reason*, *act*, and *observe* steps, and the result of each action conditions what the agent does next. The reasoning step compiles the evaluation prompt into a set of *typed physical claims* together with an executable measurement plan—the physical obligations that the video must satisfy for the prompt to hold. Each *act* step dispatches a frozen low-level physical operator (grounding and tracking with SAM 3 [7], counting over its instance identities, one of eleven typed physical measurements over those tracks, depth, optical character recognition, or audio event grounding with FlexSED [21]) or a dense-frame semantic verifier over the sampled frames. The *observe* step consumes the returned measurement and uses it to gate whether a claimed event occurred, to scope which frames and entities subsequent operators attend to, and—via fixed, deterministic rules—to compute the physical relations that the typed plan declared. Tool outputs therefore steer execution, within obligations declared before any pixel was read.

Open-ended reasoning is placed entirely at the front. The planner compiles and *statically validates* the typed measurement plan before observing the video, following plan-before-act agents [30, 70] and code-as-plan systems that emit an executable program of visual primitives [20, 63]. At execution time observations activate, skip, or localize nodes the plan already declares. Because the trajectory is fixed before execution, every measurement traces to the obligation that requested it.

Operators return *measurements*, and a deterministic composition over those measurements produces the verdict. This division — operators measure, fixed rules adjudicate — is what yields a traceable chain of evidence behind every **supported/contradicted/unknown** decision.

Compiled closures as reusable operators. The compiled closure of a typed check — its input scoping, the operator invocations it triggers, and the deterministic rule turning their measurements into a relation — is self-contained. No part of a closure is specific to one video, and each already carries the per-node cost accounting recorded in its provenance trace, so closures can in principle be named, versioned, and retrieved across tasks in the manner of procedural-memory proposals [50, 73]. PhyReAct as implemented evaluates every clip independently and retains nothing between clips; curating the closures into such a library is future work ([section 6.6](#)).

4.2 Problem formulation and system overview

A generated clip is paired with the prompt p that requested it. We represent the decoded video by its own container clock,

$$\mathcal{V} = \{(I_t, \tau_t)\}_{t \in \mathcal{T}_\mathcal{V}}, \quad \mathcal{T}_\mathcal{V} = \{0, \dots, T_\mathcal{V} - 1\}, \quad 0 \leq \tau_0 < \dots < \tau_{T_\mathcal{V}-1} \leq D_\mathcal{V}, \quad (7)$$

where I_t is frame t , τ_t is its timestamp, and $D_\mathcal{V}$ is the clip duration. The explicit range on t separates a frame index from time in seconds and also permits nonuniform decode timestamps. An audio track, when present, is addressed on the same clock.

PhyReAct has four roles. A *text-only planner* decomposes the full prompt and writes a verification program before seeing any frame. A *video-aware semantic verifier* reads timestamped frames densely for predicates such as existence and event occurrence. *Specialist instruments* return targeted measurements such as counts, tracks, masks, depths, text, and sound intervals. Finally, a *deterministic aggregator* applies typed predicates and three-valued composition. The planner and semantic verifier may use the same served checkpoint, but they are separate calls with different inputs and responsibilities.

Surface operation	Question posed	Typed check
<code>judge("assertion")</code>	dense-frame semantic verification of the complete assertion	general semantic
<code>exists("object")</code>	whether the object is visible at any decoded time	existence
<code>eq(count("phrase"), N)</code>	whether measured cardinality equals N	count
<code>before(window(A), window(B))</code>	whether event A precedes event B	temporal order
<code>during(window(A), window(B))</code>	whether A is temporally contained in B	temporal containment
<code>rel("A B", relation)</code>	a fixed-vocabulary spatial relation	spatial
<code>traj("phrase", relation, target)</code>	a fixed-vocabulary motion predicate	trajectory
<code>sound("description")</code>	whether the audio track contains the event	audio occurrence
<code>text("STRING", carrier=...)</code>	whether the named surface renders the string	rendered text

Table 5 The planner’s surface vocabulary. `count` and `window` are structural terms. The deterministic compiler expands a surface operation into its required execution closure. Thus `before` adds occurrence gates and two timing actions; `rel` adds co-visibility and mask actions and, for *behind* or *in front of*, depth actions. These dependencies need not be written as extra surface operations, but they must be present in the compiled graph. A `sound` operation reads the audio track; no audio track yields **unknown**, not a negative measurement.

The planner produces exact prompt-grounded claims and a surface plan,

$$\mathcal{C}(p) = \{c_i = (s_i, \phi_i)\}_{i=1}^{N_p}, \quad s_i \sqsubseteq p, \quad (\mathcal{C}, \tilde{\Pi}) = \Pi_\theta(p), \quad (8)$$

where s_i is a verbatim span of p and ϕ_i is the checkable assertion attached to it. The relation $s_i \sqsubseteq p$ is enforced, not inferred after execution. Types belong to the plan’s operations and checks rather than to an unverifiable free-form claim label.

Figure 13 summarizes the resulting dataflow. Planning is *global with respect to the prompt*: the naming pass sees the whole prompt and all later claim groups share that vocabulary. Execution can be *local in time*: a declared interval may gate or scope a later declared action. This is the plan–execute branch of the reason–act–observe family (section 4.1): observations activate, skip, or localize nodes already in the program, but they do not search over alternative plans or invent a new tool recipe during execution.

4.3 Prompt claims and text-only plan synthesis

Claim extraction preserves both the exact source span s_i and its normalized assertion ϕ_i . A first pass over the whole prompt assigns stable identifiers to every distinct entity and to every separately asserted event occurrence. The claims are then planned in groups of at most four. Each group sees the full prompt, the shared names, and only its assigned claims; its local identifiers are renamed mechanically before all groups are assembled. This construction prevents two groups from silently referring to the same entity by different names or to different occurrences by one name.

The surface language follows code-as-plan visual programming [20, 63], but is deliberately restricted to the operations in table 5. For claim c_i , the planner writes `check(ci, <expr>)` with one operation or a conjunction of operations. At this stage the planner has no video input: the line states what evidence would settle the claim, not whether the claim is true.

Bounded recovery. The assembled surface plan is validated as a whole because its important constraints are cross-references: every check must consume existing actions, every identifier must denote a shared entity or event, and every claim must have a check. A failing group may be asked again with its own parser or type complaints and a restatement of the format, but never with a suggested answer. After a fixed number of

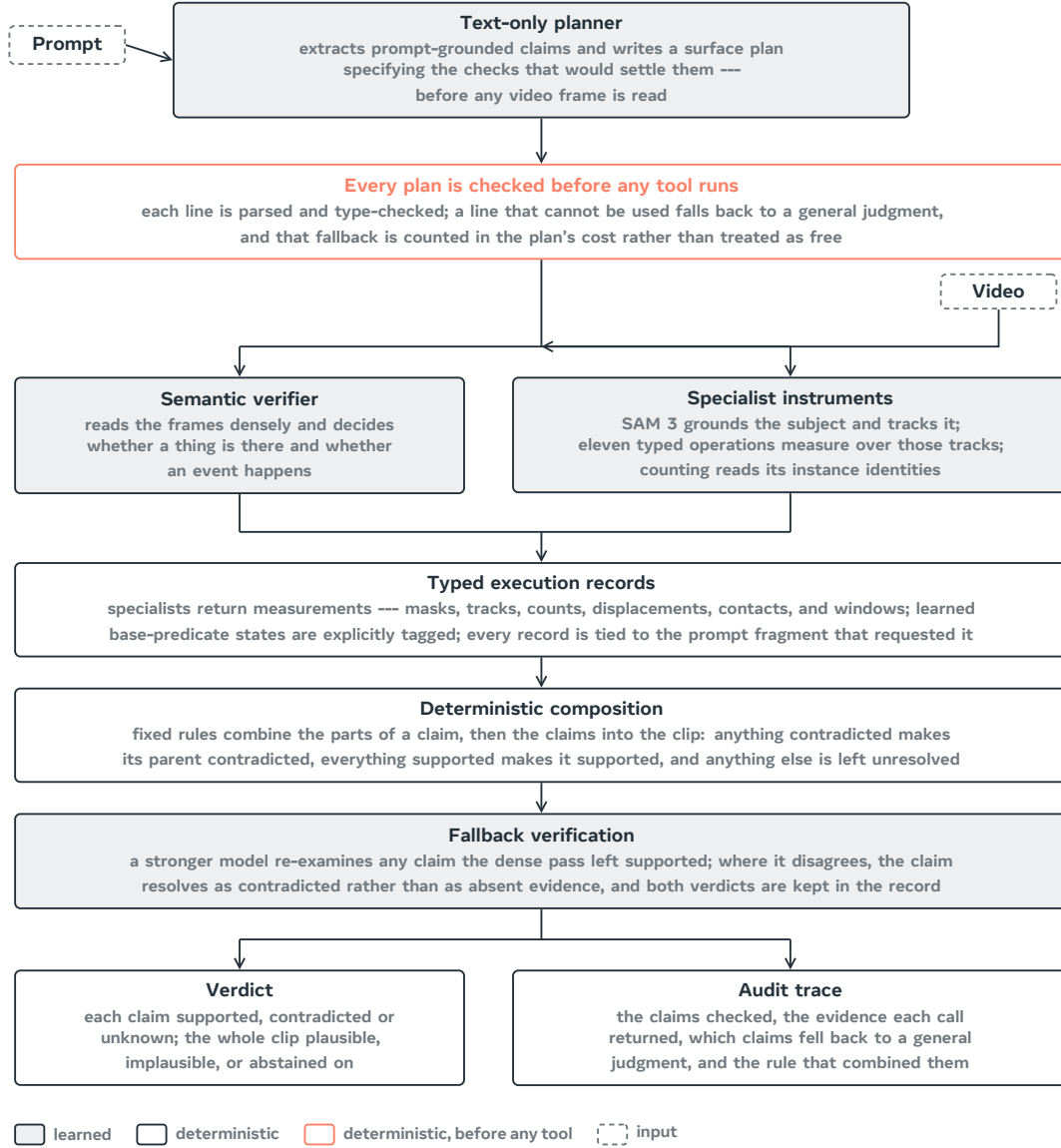


Figure 13 PhyReAct dataflow. The text-only planner receives the prompt and writes claims and a surface plan; the compiler expands and validates it before the video reaches execution. SAM 3 grounds the subject and supplies the tracks the typed operations measure over, so the specialists return numbers while the video-aware verifier returns explicitly tagged learned states. Composition is deterministic, and the one learned step after it is the fallback verifier, which may resolve a claim the dense pass had left supported. Shaded boxes are learned components; outlined boxes are deterministic.

attempts, an unusable claim falls back to direct semantic verification. The fallback is recorded and charged as a model call; it is not counted as a specialist measurement. A failed naming pass sends all claims through the same charged fallback, so plan failure never makes a prompt appear cheaper by silently dropping claims.

4.4 Typed plan representation and static validation

The compiler turns the surface program into a typed directed graph

$$G_{\Pi} = (\mathcal{A}, \mathcal{Q}, \mathcal{E}, b), \quad a_j = (o_j, \alpha_j, \omega_j, r_j), \quad b : \mathcal{Q} \rightarrow \{1, \dots, N_p\}. \quad (9)$$

Here $\mathcal{A}$ contains evidence-producing actions, including general semantic passes and specialist calls; $\mathcal{Q}$ contains typed checks; $\mathcal{E}$ points from a prerequisite to its consumer; and $b(q) = i$ binds check q to claim c_i . An action records its operation o_j , arguments α_j , declared scope policy ω_j , and result type r_j .

Before any action runs, the graph must satisfy

$$\text{Valid}(G_{\Pi}) = \text{Parse}(G_{\Pi}) \wedge \text{Refs}(G_{\Pi}) \wedge \text{Typed}(G_{\Pi}) \wedge \text{Acyclic}(G_{\Pi}) \wedge [\forall i, \exists q \in \mathcal{Q} : b(q) = i]. \quad (10)$$

Refs checks identifiers and dependency closure; Typed checks operator arity, fixed relation vocabularies, and whether every check receives the result types it requires. For example, a directed trajectory must carry a target, and a depth-ordering check must consume both masks and depth. An invalid whole line falls back to the general verifier; an invalid term can be replaced by a recorded generalist sub-check. In both cases the substitution is charged and reported rather than guessed. [Figure 14](#) shows a compiled plan of this form, with the dependency closure that [equation \(10\)](#) requires made explicit.

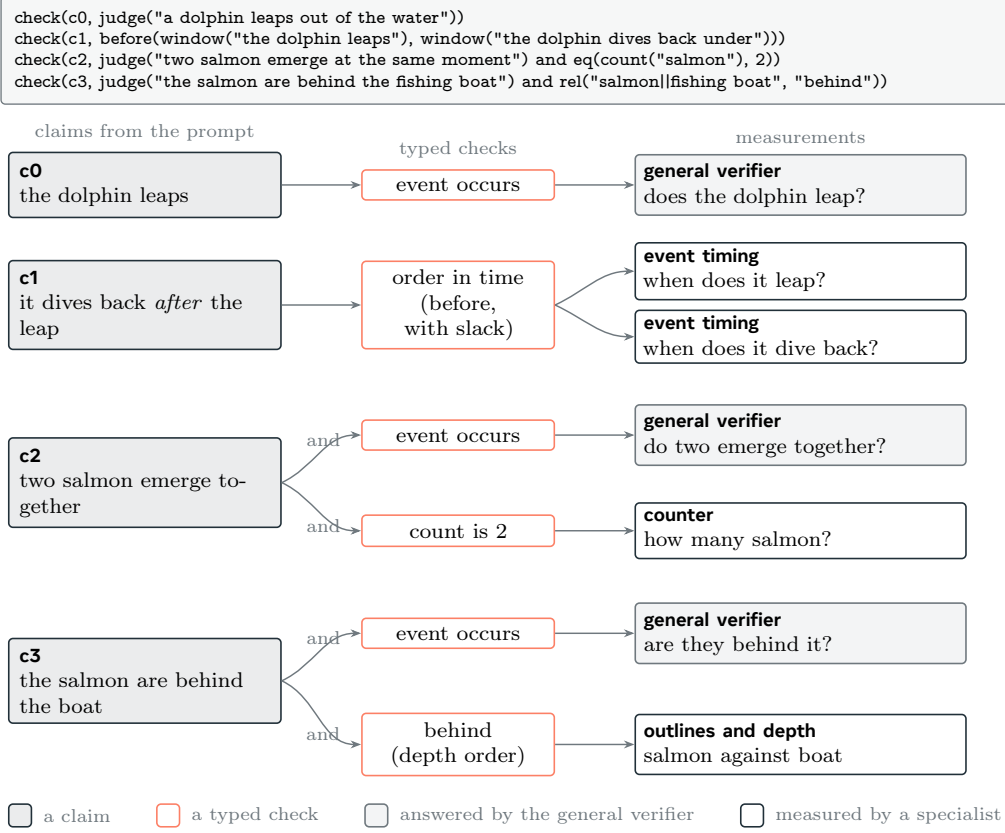


Figure 14 A surface plan after compilation into claims, typed checks, and evidence-producing actions. A numeral adds a count check beside a semantic assertion; an ordering consumes two event windows after occurrence gating; and a depth relation consumes localization, masks, and depth supplied by the compiler’s dependency closure. The graph is illustrative of the planner’s vocabulary rather than an additional learned verdict.

4.5 Timeline-conditioned multi-tool execution

Execution follows a topological order. Each action’s compiled scope function g_j converts prerequisite records into either the full frame index set or a declared local subset, after which the appropriate backend is called:

$$\Omega_j = g_j(\{R_k : (a_k, a_j) \in \mathcal{E}\}, \mathcal{T}_V), \quad R_j = F_{o_j}(\mathcal{V}|_{\Omega_j}, \alpha_j). \quad (11)$$

The scope function is part of the validated program. A usable measured window is padded, given a minimum duration, and converted to the explicit indices whose timestamps fall inside it. A missing, invalid, or overly broad window falls back to the whole clip. This conservative fallback avoids interpreting failure to localize as evidence that the requested event or relation is absent.

Occurrence gates timing. A returned time window is not evidence that the queried event occurred: a forced localizer may return its best candidate even for an absent event. Visible events are therefore confirmed first by the dense-frame semantic verifier, while audible events are confirmed on the audio track. Only supported occurrences allow their intervals to enter a temporal predicate. For a relation $R \in \{\text{before, during}\}$,

$$z_{R(A,B)} = \begin{cases} C, & C \in \{o_A, o_B\}, \\ U, & U \in \{o_A, o_B\} \text{ or } W_A \text{ or } W_B \text{ is unavailable,} \\ \rho_R(W_A, W_B), & o_A = o_B = S, \end{cases} \quad (12)$$

where o_A, o_B are occurrence states and W_A, W_B are measured windows. Thus timing may compose an order but can never establish occurrence.

the plan line the planner wrote, before any frame is read

```
check(c1, before(window("the first domino tips forward"),
  window("the first domino triggers a chain reaction")))
```

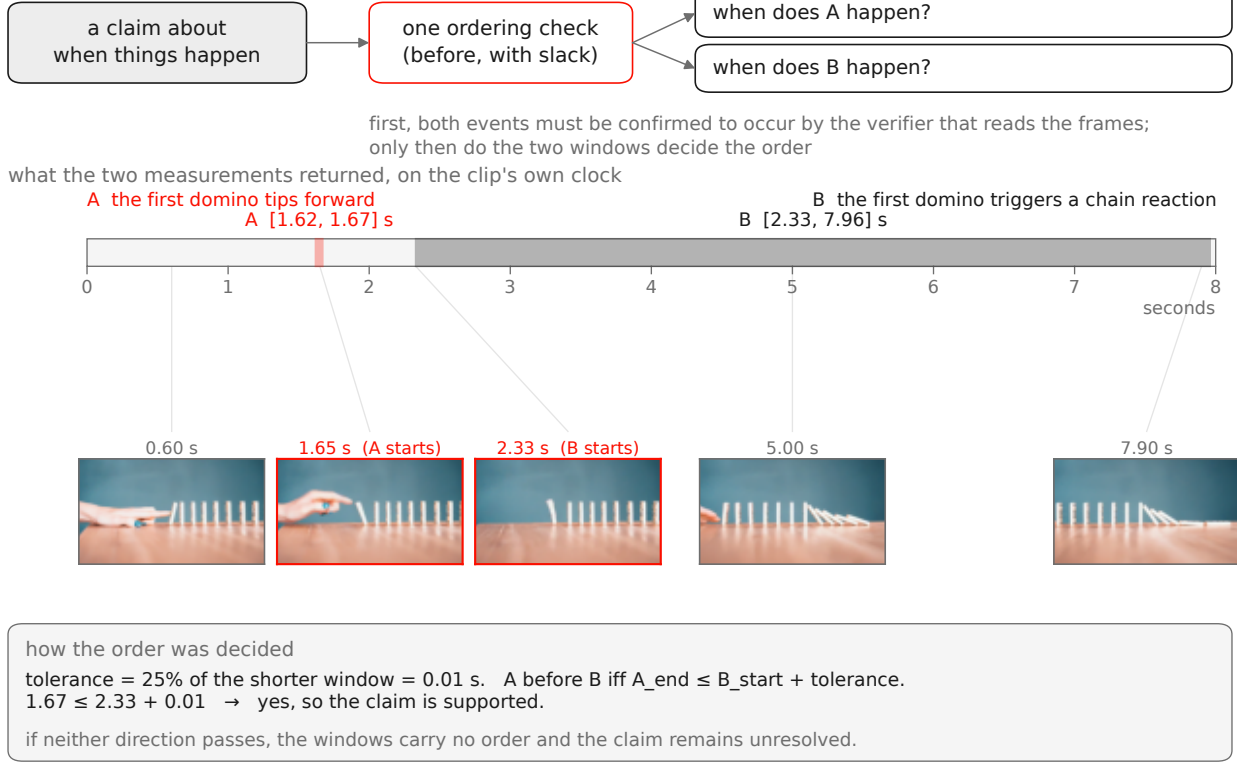


Figure 15 A real temporal trace. The planner writes the line before reading any frame. The verifier first confirms both events, the timing actions then return $W_A = [1.62, 1.67]$ s and $W_B = [2.33, 7.96]$ s, and [equation \(13\)](#) resolves their order. Every number and displayed frame comes from the recorded run.

Temporal arithmetic. For positive-duration windows $W_X = [s_X, e_X]$, the recorded executor uses a boundary slack equal to one quarter of the shorter window:

$$\delta(A, B) = \frac{1}{4} \min(e_A - s_A, e_B - s_B),$$

$$\rho_{\text{before}}(W_A, W_B) = \begin{cases} S, & e_A \leq s_B + \delta(A, B), \\ C, & e_B \leq s_A + \delta(A, B), \\ U, & \text{otherwise.} \end{cases} \quad (13)$$

The slack absorbs small endpoint uncertainty, so it can tolerate a boundary overlap no larger than δ ; if neither direction passes, the intervals do not resolve an order and the state is U. With the quarter-window rule and positive durations, the first two cases cannot both hold. Containment is directional and uses the same boundary slack: clear containment supports, clear reverse containment contradicts, and boundary-indistinguishable cases remain unknown. These predicates are a tolerance-scaled subset of Allen's interval algebra [1]. [Figure 15](#) follows one such ordering through a recorded run, from the plan line written before any frame was read to the two measured windows that settle it.

Exact-request reuse. Two actions share one execution only when their complete realized signatures match:

$$\chi_V(a_j) = (\text{id}(\mathcal{V}), o_j, \alpha_j, r_j, \Omega_j), \quad a_j \sim a_k \iff \chi_V(a_j) = \chi_V(a_k). \quad (14)$$

The executor calls each equivalence class once, following exact common-subexpression reuse [54, 57]. Because the realized scope Ω_j is part of the signature, whole-clip and localized requests never merge merely because

Capability	Running backend	Returned object
Semantic predicate	dense timestamped Qwen3-VL pass [51]	learned base state with localized rationale
Fallback verification	gpt-5.6-sol, invoked on a claim the dense pass passes	three-valued claim state, both verdicts recorded
Grounding and masks	SAM 3 Video [7] with persistent object identities	per-frame masks and tracked identities
Object count	cardinality of SAM 3 instance identities	count with its per-frame series
Physical measurement	eleven typed operations over the tracks: motion, contact, shape, hops, path, presence, damping, onsets, events, timing, viewpoint	typed numeric measurement with its evidence
Camera cancellation	dense optical flow subtracted before subject motion is reported	subject-relative displacement
Event timing	Qwen3-VL harness over timestamped frames	interval, absence, or refusal
Rendered text	PaddleOCR recognition head, every frame, EasyOCR fallback	literal glyph transcription
Audio	FlexSED sound-event detection [21]	sound intervals on the audio track

Table 6 Execution backends and their outputs. SAM 3 grounds the subject and supplies the tracks every physical measurement is taken over, and counting reads the cardinality of its instance identities. The eleven typed operations return numbers with their evidence; the two learned components are identified as such, and the fallback verifier is the only one that may resolve a claim. Displacements are expressed in multiples of the subject’s own radius, which requires no scene calibration, and camera motion is cancelled before any subject motion is reported. No holistic video-quality model produces the final clip decision.

their natural-language queries are similar.

Execution backends. Table 6 lists the backend running behind each capability in the evaluated configuration. Three sub-checks remain learned, resisting full symbolization: subtype identity, dense action-event recall, and depth-ordered spatial interpretation on localized evidence. Their outputs carry a learned base-predicate tag, distinguishing them from deterministic consequences of a measurement. No holistic scorer appears in the verdict path.

4.6 The physical measurement operations

Physical claims are settled by eleven operations sharing a single grounding step. SAM 3 segments the named subject and carries a persistent identity for it across frames, and every operation measures over those tracks. Each operation corresponds to a class of human complaint in the corpus.

The largest class is displacement, because the dominant physical complaint is one sentence in many costumes: the sun remains stationary instead of rising, the tornado stays in place, the plate never rotates, the clouds are static. Something the prompt sets in motion does not move, or moves the wrong way. Complaints resembling a ballistic arc are rare by comparison. Three conventions make that measurable. Displacement is reported in multiples of the subject’s own radius, so no scene calibration is required and a sun crossing a tenth of its own width is distinguishable from one crossing three. Direction is taken from the furthest excursion off the starting position rather than the net, because a rise followed by a fall cancels to nothing. And camera motion is cancelled by dense optical flow over the whole frame before any subject motion is reported, which requires no stationary landmark and so remains valid when the camera and the subject move together.

Three operations address failures that involve no net travel. Contact measures the closest approach between two tracked masks, which decides whether two things touch at a finer resolution than box overlap. Shape change measures the subject’s own surface motion relative to the background flow, catching a scroll that

rolls when it should lie still — a case displacement reports as zero. Path shape reads the trajectory’s form, distinguishing a circle from a figure of eight.

The rest address time and existence. Permanence asks whether a subject that should vanish instead persists, and hop counting whether one that should leave the ground ever does. Damping detects a spinning object that never slows, the complaint about the arrow of time. Onset ordering decides which of two events comes first, and covers the more common case in which one of them never happens at all. Clip timing handles duration and playback rate, and viewpoint motion separates a moving camera from a moving subject when the claim concerns the camera.

All eleven return a number with the evidence behind it — the frames read, the tracker used, the threshold applied — and leave the decision to the fixed rules of [section 4.8](#). All eleven may also abstain: a subject smaller than one percent of the frame, or a track too short to measure, yields an explicit abstention, so an unmeasurable claim does not resolve as a supported one.

4.7 Evidence records and provenance

Each action is serialized as a claim-bound record

$$R_{j,i} = (\text{id}(\mathcal{V}), p, s_i, o_j, \Omega_j, \sigma_j, \mu_j, h_j, \kappa_j), \quad \sigma_j \in \{\text{measured}, \text{abstained}, \text{errored}\}. \quad (15)$$

The payload μ_j contains either a typed measurement or an explicitly tagged learned base state; h_j contains the native-output and artifact hashes; and κ_j contains calls and accelerator time, with optional wall time, price, and token counts. Shared execution may bind the same record to several claims, but its cost is charged once by execution identifier.

The three statuses are mutually exclusive. A **measured** record carries a payload and no abstention or error; **abstained** carries a reason and no payload; **errored** carries an infrastructure message. A valid negative measurement—for example a complete usable track that does not exhibit the requested motion—remains **measured** and may contradict a claim. Failure to track because of blur, occlusion, low quality, or an instrument refusal is **abstained**, not a negative observation.

The counting path violates that boundary. Its contract reduces a per-frame series to a single number by taking the mode, so a phrase the segmenter grounds intermittently yields a series such as $[4, 6, 0, 6, 6, 0, 0, 0, \dots]$ whose mode is 0, and a grounding that failed on most frames is reported as a measured count of zero. On a clip containing five steel balls the operation returns exactly that: 0 over 241 frames read, accompanied by **stable**: **false** and a range of 0 to 8. A consumer reading the scalar alone therefore receives a confident wrong number, and the auxiliary fields are load-bearing. In the present payload a zero meaning *absent* is indistinguishable from a zero meaning *never found*; [section 9](#) gives the consequence for count-derived figures. A terminal **errored** record is retried and, if still terminal, produces no semantic verdict; the clip is marked $\perp_{\text{infra}}$ and excluded from the completed evaluation set rather than relabeled **unknown**.

Specialist schemas reject judgment-shaped fields, nonfinite numbers, duplicate keys, and noncanonical payloads. Every record carries the full prompt and the verbatim source span s_i , and heavy artifacts are content-addressed by path and byte hash. The schema thus guarantees two things: a specialist cannot emit the final prompt-satisfaction label, and accepted evidence is attributable and replayable. It cannot make a learned measurement unbiased, nor detect a well-formed payload that is wrong.

4.8 Three-valued verification and aggregation

For a completed trace, every sub-check resolves in

$$\mathbb{V} = \{\text{S}, \text{C}, \text{U}\},$$

standing for **supported**, **contradicted**, and **unknown**. A typed resolver ψ_q maps measured parent payloads to a sub-check state; it is deterministic for count, timing, track, transcription, and other explicit predicates, and is

learned only for the semantic sub-checks named above:

$$z_q = \begin{cases} \psi_q(\{\mu_j : a_j \in \text{Pa}(q)\}), & \sigma_j = \text{measured for all required } a_j, \\ \text{U}, & \text{a required action abstained or produced no usable evidence,} \\ \perp_{\text{infra}}, & \text{a required action has a terminal infrastructure error.} \end{cases} \quad (16)$$

Learned semantic checks use an asymmetric gate: they may contradict only when a positively contradicting feature is visible; unresolved visual evidence, unparseable output, or missing modality yields U. This avoids turning failure of an instrument into an accusation about the video.

Negation is applied to the positive sub-check state, never by asking a tool to measure a negative quantity:

$$\neg_3 \text{S} = \text{C}, \quad \neg_3 \text{C} = \text{S}, \quad \neg_3 \text{U} = \text{U}. \quad (17)$$

Let $K_i = \{q \in \mathcal{Q} : b(q) = i\}$ be the validated checks for claim c_i , after applying [equation \(17\)](#) where required. Claim and clip composition are

$$y_i = \begin{cases} \text{C}, & \exists q \in K_i : z_q = \text{C}, \\ \text{S}, & \forall q \in K_i : z_q = \text{S}, \\ \text{U}, & \text{otherwise,} \end{cases} \quad Y = \begin{cases} \perp_{\text{infra}}, & \text{the trace terminates with an infrastructure error,} \\ \text{IMPLAUSIBLE}, & \exists i : y_i = \text{C}, \\ \text{PLAUSIBLE}, & N_p \geq 1 \wedge \forall i : y_i = \text{S}, \\ \text{ABSTAIN}, & \text{otherwise.} \end{cases} \quad (18)$$

The cases are ordered: one contradiction gives an implausible clip, plausible requires every planned claim supported, and a trace with no contradiction but at least one unknown abstains. PLAUSIBLE is therefore a statement about the claims the plan executed.

Conflict resolution between two verifiers. A claim the video-aware verifier passes may be re-examined by a stronger reviewing model, and that model’s verdict stands: where it overrules a pass, the claim resolves to **contradicted** rather than to **unknown**. Formally, for a claim whose sub-checks would roll up to S under [equation \(18\)](#), the reviewer’s state replaces y_i . Both verdicts are kept in the record, so a row can be re-scored on the first verifier’s alone.

The direction is asymmetric by design. The reviewer examines only claims the first verifier passed, and never a specialist measurement, because a measurement carries evidence that a second opinion has no way to inspect.

Reading on-screen text. A recognition engine reads frames directly to settle text claims. Asked what a sign says, a generative model repairs what it sees, rendering a malformed **Stly your** as the well-formed **Style your** the prompt requested and concealing the defect under test; a recognition head returns the malformed string, which a deterministic comparison rejects. The engine reads every frame rather than a sampled fraction, since a caption present for part of a clip can fall between samples, and each record states the frames read.

4.9 Diagnostic interface: the contradiction packet

Beyond the clip state Y , a run emits a localized contradiction packet

$$\mathcal{F}(p, \mathcal{V}) = \{(c_i, \{R_{j,i}\}_j, \Omega_i, B_i) : y_i = \text{C}\}, \quad (19)$$

where Ω_i is the union of available temporal support and B_i contains available box tracks or masks. The packet is assembled from the trace that produced the verdict, so each contradicted claim carries both the evidence records behind the conclusion and the region of the clip in which the failure occurs: *what* failed, *on what evidence*, and *where*. That space–time localization is what [Section 6](#) maps onto a local edit of the generator control ([Section 6.4](#), (27)).

Closed-loop refinement is not run in this report. The critic and the simulation-conditioned generator are evaluated on disjoint data, and no **PhyReAct** packet has yet been used for preference optimization, denoising guidance, candidate selection, or localized regeneration. [Equation \(19\)](#) defines the interface only; we make no claim that the current generator has been optimized by the critic or that any packet has made a generated clip more physically correct.

5 Evaluation Protocol and Measurement

Target and reference standard. The evaluation core of [section 3](#) contains 149 clips carrying 304 annotated flaws. These flaws are the reference standard, under the conventions of [section 3.1](#), and the denominator for all core recall analyses unless stated otherwise; the evaluation measures agreement with the annotated flaws rather than detection of every defect present in a clip. Because these clips were inspected during system development, the core results describe performance on development data; the held-out evaluation is reported in [section 5.7](#). Dashes in tables are defined in each table’s caption.

Implementation. Qwen3-VL-30B-A3B-Instruct [51], served in BF16 without quantization and with tensor parallelism across two GPUs, is used for both the text-only planner and the video-aware semantic verifier. These are separate calls: the planner receives the prompt and claim state but no frames, while the verifier receives densely decoded, timestamped frames and a single declared predicate. The evaluated run uses frozen specialist revisions and the backends listed in [table 6](#).

Dense-frame requirement. All evaluated video-aware components receive video through the logged dense-input path. Individual measurements may operate on a recorded subset of the decoded frames; measurements produced from a sparse-input path are excluded. A rerun is included only if it uses this dense input path, the same frozen evaluation target, BF16 model inference, and pinned tool revisions.

Cost and uncertainty. A detector call, a tracker call and a full vision-language judgement do not cost the same, so cost is reported as model and tool calls per clip rather than as wall-clock time. Uncertainty is estimated by resampling whole clips, because two flaws in one clip are not independent.

Planner validity. Before end-to-end evaluation, the prompt-only planning stage was inspected without video or tool access. For 233 claims from 22 annotated clips, the planner produced a valid, compilable plan for every claim, using one or two planner calls per clip; 27 claims (11.6%) contained at least one specialist action that survived validation, compared with 2.3% under an earlier tabular plan format.

Diagnostic routing coverage. On diagnostic sets composed of claims that require quantitative measurement, the routing rate was 19/30 pooled across counting, spatial, and ordering claims, and 10/10 for synthetic counting prompts. These diagnostics test routing and syntax; they are not estimates of verdict accuracy or coverage on the evaluation core.

Unsupported predicates and fallback handling. The supported predicate vocabulary lacks an “on top of” spatial predicate and rejects negated quantitative measurements; claims using these forms are routed to the generalist fallback verifier and counted as fallback cases.

5.1 Matching critic findings to annotated flaws

Matching is performed after critic inference and cannot affect the critic’s outputs. The unit of comparison is a pair of sentences: what the annotator wrote about the clip, and what the critic reports the video showed. Comparing the prompt fragment against the critic’s restatement of it would instead measure agreement about the prompt, since both are assertions of the same requirement.

Each pair is judged on meaning and assigned one of three outcomes. A *whole* match requires the two statements to assert the same defect: different wording, a different name for the same object, and a measured quantity in place of a description all qualify, while two different problems with the same object do not. Failing that, a *partial* match requires the critic’s finding to identify at least one aspect of a compound annotated defect, which an annotator’s single sentence can span while the critic reports one aspect at a time. Pairs meeting neither condition are non-matches.

Whole and partial pairings then form the candidate edges for maximum bipartite matching, subject to the constraint that each finding and each annotated flaw may be used at most once; after assignment, each flaw is reported as whole, partial, or missed. Greedy assignment can reduce the number of matched flaws: if one flaw

Annotated flaw category	Annotated	Whole	Partial	Found	Recall
action or event occurs	137	100	14	114	83%
how many	37	10	8	18	49%
object identity	30	14	6	20	67%
on-screen text	25	21	0	21	84%
spatial relation	18	10	1	11	61%
order and timing	17	10	3	13	76%
attribute	16	10	2	12	75%
camera and style	10	7	2	9	90%
other	7	5	1	6	86%
physical motion	7	4	0	4	57%
All	304	191	37	228	75.0%

Table 7 Flaw recall on the evaluation core, by annotated flaw category: 228 of 304 human-annotated flaws over 149 clips. Whole and partial are the match outcomes defined in [section 5.1](#); Found is their sum. Categories are the annotators’ labels, assigned before any critic ran.

can be matched by either of two findings and a second flaw can be matched by only one of them, assigning that finding to the first flaw leaves the second unmatched. The matching judge is distinct from all evaluated system components. For each pair, it initially produces three votes at temperature zero. Each unparsable reply is retried once; if the retry remains unparsable, that vote is recorded as a matching error. Among valid votes, disagreement is resolved by assigning non-match if any valid vote is a non-match, otherwise partial if any valid vote is partial, and otherwise whole. If no valid vote remains after retries, the judge produces nine new votes and assigns the majority outcome. A run aborts rather than reporting a number if the matching judge cannot produce a usable matching outcome for more than 2% of annotated flaws.

5.2 Flaw recall on the evaluation core

Across the 149 clips of the evaluation core, which carry 304 human-annotated flaws, PhyReAct finds **228 of them (75.0%)** under the protocol of [section 5.1](#). Of these 228 flaws, 191 are whole matches, each covered completely by a single finding; the remaining 37 are partial matches, each consisting of a finding that addresses an aspect of the annotated defect without covering it completely. On 91 of the 149 clips, the critic finds every annotated flaw in that clip.

Two components contribute to matched findings in distinct ways. For 83 of the 228 found flaws, the fallback verifier, gpt-5.6-sol, overrode a pass from the dense semantic verifier and changed the claim state to contradiction. Separately, the on-screen-text recognizer transcribes malformed glyphs literally rather than normalizing them to the requested phrase, allowing string comparison to identify garbled text.

Broken down by the annotators’ own categories ([table 7](#)), recall runs from 90% to 49%. Camera and style (90%) and on-screen text (84%) have the highest observed recall: both are evaluated using a direct measurement against an explicit target. Counting has the lowest recall at 49%: as described in [section 4.7](#), the counting path reduces a per-frame series to its mode, so a subject detected intermittently by the segmenter may be reported as absent. The physical-motion category contains seven flaws, four of which are found. Most of these seven flaws concern deformation — cloth that folds wrongly or a body that bends where it should not; such defects lie outside the trajectory measurements, and given the category’s size, its recall is interpreted descriptively.

5.3 Recall against clip properties

Category-level recall describes the kinds of defects the critic identifies. [Table 8](#) instead groups the same 304 flaws by four clip properties: annotator severity, clip duration, number of annotated flaws, and difficulty stratum.

Group	Annotated	Found	Recall
<i>Annotator severity</i>			
severity 1	3	3	100%
severity 2	28	22	79%
severity 3	146	107	73%
severity 4	99	78	79%
severity 5	28	18	64%
<i>Clip duration</i>			
6 s or less	76	60	79%
6-8 s	46	29	63%
over 8 s	182	139	76%
<i>Number of annotated flaws per clip</i>			
1	63	49	78%
2	92	65	71%
3-4	113	87	77%
5 or more	36	27	75%
<i>Difficulty stratum</i>			
hard	83	60	72%
not hard	221	168	76%

Table 8 Recall grouped by annotator severity, clip duration, number of annotated flaws in the clip, and difficulty stratum, on the run of [table 7](#). Severity is the annotator’s own 1–5 judgement of how badly the clip fails.

Across the four groupings, subgroup recall remains within approximately ten percentage points of overall recall, a range comparable to the variation observed when the matching judge reevaluates unchanged text. Severity 5 has the lowest recall, at 18 of 28 flaws. These descriptive estimates show no clear association of either annotator severity or difficulty stratum with whether the critic finds a defect.

5.4 Illustrative specialist measurement traces

[Figure 16](#) presents six illustrative matched findings from an earlier physics-routing pass over the evaluation core; they are not drawn from the run of [table 7](#) and do not contribute to its recall. Each panel shows the clip, the annotator’s sentence, the specialist evidence, and the resulting verdict. Five cases rest on numeric measurements; the text case rests on a recognition-head transcript compared as a string.

Case selection. The selection rule was fixed in advance. Cases were selected from clips for which the matching protocol linked every annotated flaw to a critic finding; a case was included only when the raw per-check record attributed the finding for that flaw to a specialist measurement and recorded the measurement values. Ten candidates were excluded: two because the protocol linked a different flaw from the one addressed by the tool, seven because the video-aware verifier produced the finding while the specialist abstained, and one because the clip-timing operation misread the aspect ratio “vertical 9:16” as a duration comparison of 9.0 against 16.0 seconds, a finding with the correct conclusion but an invalid rationale.

What each measurement discriminates.

- *Dolphin.* Asserted: a leap. Measured: vertical spread, 0.65 animal radii; horizontal spread, 0.20 radii; largest single excursion from the animal’s baseline, 0.00 radii. Comparator: 0.30 radii for leaving the water. Verdict: contradicted, although a motion-only test would have passed the clip.
- *Bottle.* Asserted: the bottle rolls. Measured: 10.20 px/frame surface motion against 11.00 px/frame background motion. Comparator: comparable magnitudes indicate camera rather than object motion. Verdict: contradicted, although the scene contains substantial apparent motion.

cfdfe797 **leap detector**



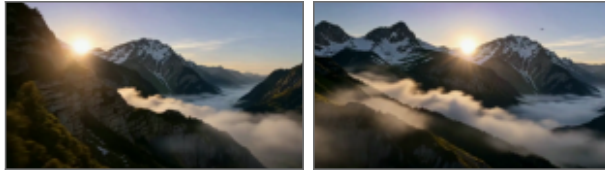
first frame 0.00 s last frame 5.00 s
 annotator The dolphin ... merely swims to the surface and then dives back down without jumping out of the water.
 measured hops: required yes, measured no – largest vertical excursion 0.00 of the dolphin's own radius over 37 tracked frames; 0 excursions reach the 0.30-radius threshold
 verdict **contradicted**

f85ffa1f **surface motion vs. background flow**



first frame 0.00 s last frame 10.24 s
 annotator The water bottle ... remains in its original position and never rolls on the floor.
 measured rolling: required yes, measured no – surface moves 10.20 px/frame against 11.00 px/frame of background flow, over 256 frame pairs
 verdict **contradicted**

a0b49065 **vertical travel, camera cancelled**



first frame 0.00 s last frame 8.00 s
 annotator The sun remains completely stationary in its original position without slowly rising.
 measured vertical travel: required up – largest departure from its starting height 0.71 radius, under the 0.75 that counts as travel; net -2 px over 192 tracked frames
 verdict **contradicted**

128c7e8d **counter**



first frame 0.00 s last frame 8.00 s
 annotator There are only five babies in the video, which does not match the quantity requested in the text prompt.
 measured count: prompt asserts 7, measured 5 (exact mode)
 verdict **contradicted**

18a38392 **counter**



first frame 0.00 s last frame 5.00 s
 annotator A total of four fighter jets appear in the video, which does not match the requested quantity of three.
 measured count: prompt asserts 3, measured 4 (exact mode)
 verdict **contradicted**

4075411a **OCR recognition head**



first frame 0.00 s last frame 10.00 s
 annotator The text appearing in the video does not match the requested text "Congratulations to the graduates".
 measured text: required 'CONGRATULATIONS GRADUATES', read 'GOBADRAEAT (CRSTRORISEUND' – stable wrong transcript on 25 of 241 frames (stride 10)
 verdict **contradicted**

Figure 16 Six flaws identified through specialist measurement. Each panel gives the clip identifier and the specialist that answered, the first and last decoded frames of the clip, the human annotator’s sentence verbatim, the specialist’s own required-versus-measured pair with its evidence, and the resulting claim state. Displacements are normalized by the subject’s radius, so no scene calibration is required. In the text case, the boxed banner region of that last frame is magnified in place, showing the render’s own glyphs. Measurement text is copied from the run’s records.

Judge	All clips ($n = 200$)				Hard subset ($n = 55$)	
	Exact	Within 1	PLCC	SRCC	PLCC	SRCC
VideoScore2 [23]	33.0%	83.0%	0.353	0.327	0.239	0.151
PhyReAct	31.0%	76.0%	0.464	0.465	0.480	0.460

Table 9 Semantic-adherence agreement with human ratings on the 200-clip VideoPhy-2 sample described in the text, scored with VideoScore2’s evaluation script. The physical-commonsense dimension is excluded because the critic does not emit a score on that scale.

- *Sun*. Asserted: a rise. Measured: peak departure from the starting height, 0.71 radii, used instead of net displacement because a rise followed by a fall cancels. Comparator: 0.75 radii, with 0.67 radii of camera motion over the same interval. Verdict: contradicted.
- *Counting (two cases)*. Asserted counts compared exactly against measured counts.
- *Text*. Asserted: the requested phrase. Measured: the recognition head returns `GOBADRAEAT(CRSTRORISEUND`. Comparator: string equality, which fails; a generative model asked the same question normalizes the string to the requested phrase.

Independent measurements on one clip. The dolphin clip carries a second annotator sentence stating that the animal never disappears into the depths; two further specialists independently evaluated the corresponding prompt requirement that the animal disappear. The vertical-travel check required downward motion but measured a maximum displacement of 2.07 radii upward, corresponding to a net image-coordinate displacement of -317 px over 37 tracked frames, where negative vertical displacement denotes upward motion. The persistence check required the dolphin to vanish and found it on 121 of 121 frames, with no gap of three frames or more. Both specialist measurements independently contradict that prompt requirement.

Measurement coverage. The record also states the temporal coverage of each measurement. In the banner case the recognition head read 25 of that clip’s 241 frames at a stride of ten, and the record states both figures. A transcript stable across those 25 frames is strong evidence about a banner present throughout the clip; a caption appearing only briefly could fall between the frames sampled.

Specialist abstention. In the evaluated run of [table 7](#), physical specialists were called on 227 checks, abstaining on 166 and contradicting the prompt on 61; each abstention carries the reason it occurred. The most common abstention occurred when no candidate grounding passed screening: the segmenter produced no usable track for the named subject. Groundings are screened by a semantic guard, shown one frame with the candidate boxes drawn on it and asked only whether those regions correspond to the named subject. On a clip requesting seagulls scattering in the wake of fighter jets, the boxes offered for “the seagulls” contained the jets, and the guard discarded them. The check then yields **unknown**, separating a failure to find the subject from a violation by it.

5.5 Agreement with human ratings

Because localized flaw recall and clip-level scalar scores are not directly comparable, this subsection compares PhyReAct and VideoScore2 [23] using human clip-level semantic-adherence ratings. VideoScore2 produces ratings on the same 1–5 scale, and the analysis follows its published evaluation script. The analysis uses a fixed-seed uniform sample of 200 VideoPhy-2 [4] test clips, 55 of which carry the benchmark’s *hard* flag; the sample, score definition, and metrics were prespecified, and the scoring function was not fitted to these clips. Each judge assigns semantic adherence an integer rating from 1 to 5. Agreement with the human rating is measured by exact accuracy, accuracy within one point, PLCC, and SRCC; for each metric, clips with an unparsed prediction from either judge are excluded.

As [table 9](#) shows, VideoScore2 has higher exact and within-one-point agreement, whereas the critic has higher PLCC and SRCC. In a paired bootstrap over the same 200 clips, the SRCC difference is $+0.138$ with a 95% interval of $[+0.005, +0.277]$; the PLCC difference is $+0.110$ with an interval of $[-0.018, +0.242]$. On

the 55 hard-flagged clips, the differences are +0.242 for PLCC ([+0.018, +0.511]) and +0.309 for SRCC ([+0.050, +0.583]). This pattern is consistent with evidence assembled per claim degrading more gracefully on difficult clips than a single learned scalar, but the hard subset is small and the intervals are wide.

Two additional limits apply. The sample is a single fixed draw of 200 clips, so the estimates describe agreement on that sample rather than across the benchmark. The two judges are also not cost-matched: the critic runs a decomposition and a set of tool calls per clip while VideoScore2 runs a single forward pass, so this comparison carries no information about efficiency.

5.6 Comparison against published evaluators

This subsection compares PhyReAct with evaluators whose outputs can be linked to specific defects, and separately reports whether CLIPScore tracks aggregate clip properties. Question-decomposition and scene-graph methods produce per-question answers that can be linked to the generating claims, while free-text evaluators produce localized but unstructured flaw lists; modular video question answering is listed for context but was not evaluated under this protocol. Clip-level agreement with human ratings is addressed in [table 9](#).

The same served Qwen3-VL-30B-A3B-Instruct model was also evaluated in a single pass, given the prompt and every frame of the clip in one call, without planning, tools, or per-claim verification. Under the matching protocol its free-text findings match 222 of the 304 annotated flaws, at one model call per clip. [Table 10](#) reports this result separately because free-text flaw lists are prompt-sensitive and do not provide the per-verdict evidence produced by the framework; recall alone does not capture that difference.

Question decomposition matches **164 of the 304 annotated flaws (53.9%)**: 125 whole and 39 partial matches, against PhyReAct’s 228 ([table 10](#)). Both evaluators read the same 149 clips densely and send their questions to the same served model; for 148 clips they also use identical claim lists. Question decomposition asks one closed question per claim and accepts the answer, whereas PhyReAct plans measurements and may apply multiple checks to a claim.

PhyReAct’s higher recall does not correspond to a higher match rate per reported finding: question decomposition matches a human flaw for 29% of its findings, compared with 26% for PhyReAct. It reports substantially fewer findings — 557 compared with 871 — and returns a passing answer for most questions. The question forms help explain this pattern. Roughly one third ask only whether an object is present, for example, *Is there a vending machine?*; such questions can often be answered from one frame, and static objects are comparatively easy for the generator to render. Twelve clips receive no finding at all. Clip by clip, question decomposition finds fewer flaws than PhyReAct on 70, the same number on 63, and more on 16.

As an aggregate clip-level diagnostic, CLIPScore correlates, across all 149 clips, with severity at $r = +0.035$ ($p = 0.67$) and with annotated-flaw count at $r = -0.024$ ($p = 0.77$). Because 86 of the 149 clips contain more than one annotated flaw, accounting for 241 of the 304, a single clip-level score cannot identify individual flaws.

5.7 Held-out evaluation on Physion-Eval

Physion-Eval ([section 3.5](#)) provides clips that were not used in the development of this system. Its public release contains 9,569 generated clips from five current generators, each accompanied by a human report describing the clip’s physical glitches; 8,745 clips have at least one reported glitch. Two properties of the release bound what can be measured on it. The matched real-world reference videos required for the benchmark’s J statistic are withheld for copyright, and clips without reported glitches are not verified negatives. The public release therefore supports neither J nor false-alarm rates, for any evaluator. It supports evaluator flag rates on release clips and matched-report rates on clips with a reported glitch.

PhyReAct is being run over a fixed randomized stream of the release; the figures are interim and cover all of the first 601 clips scored: 543 with a reported glitch and 58 without. The in-context lesson state of [section 6](#) was active and updated as the stream progressed, so the figures describe this evolving configuration rather than a single frozen system; the run is prequential, and no component saw a Physion-Eval label before its clip was scored.

Evaluator	Output it produces	Found	Recall	Mean model/tool calls per clip
PhyReAct	localized flaw finding	228	75.0%	14.1
<i>Other evaluators with structured localized outputs</i>				
Question decomposition [26]	per-question answers	164	53.9%	21.9
Davidsonian scene graph [9]	per-question answers, dependency-gated	161	53.0%	52.1
Modular video QA [12, 46]	per-question answers	—	—	—
<i>Free-text localized output</i>				
Single-pass VLM, same model [51]	free-text flaw list	222	73.0%	1.0

Table 10 Localized flaw recall over the same 149 clips and the same 304 flaws, judged by the protocol of [section 5.1](#). The calls column reports the mean number of model and tool calls per clip and excludes calls made by the matching protocol, which scores outputs rather than generating evaluations. Question decomposition uses the same claim list as PhyReAct on 148 of the 149 clips. The free-text row uses the same served model as PhyReAct, prompted once with every frame; it is listed separately because it produces an unstructured flaw list rather than structured per-question answers or per-verdict evidence. A dash marks an evaluator not evaluated under this protocol.

To distinguish prompt-grounded findings from physics-grounded benchmark reports, three measures are reported, defined below and labeled in [table 11](#). The *matched-report clip rate* is the share of reported-glitch clips for which at least one critic finding matches the annotator’s report under [section 5.1](#). The *evidence-conditioned flag rate* is the share of clips classified as physically unrealistic by the served model using only the records of every check — the question, verdict, stated reason, and numerical evidence — without receiving frames or having to identify the reported defect. The *anomaly-phrased finding rate* is the share of clips with at least one finding that explicitly alleges a physical anomaly rather than an unmet prompt requirement.

[Table 11](#) reports the three rates alongside the benchmark’s published J and flag-rate results; because this work’s rows and the benchmark’s rows use different samples and system configurations, their juxtaposition is descriptive rather than a common-protocol comparison.

The evidence-conditioned flag rate is nearly twice the anomaly-phrased finding rate, indicating that the accumulated evidence supports physical-implausibility judgements beyond what the wording of individual findings states.

5.8 Scope and interpretation

The evaluations use distinct settings and should be interpreted separately. Core recall uses recorded annotations on development data, where an unrecorded defect cannot be scored either way; the question-decomposition baseline was specified after inspection of critic outputs; the VideoPhy-2 analysis measures clip-level agreement with human ratings; and the interim, prequential Physion-Eval evaluation uses held-out clips with an evolving lesson state, a different annotation vocabulary, and no use of the frozen core protocol.

Evaluator	J exo	J ego	Flag rate
Untrained humans (reference)	30.7%	56.0%	52.7%
<i>This work, judging from the critic’s measured evidence</i>			
Evidence-conditioned flag rate	—	—	26.8%
Matched-report clip rate	—	—	18.4%
Anomaly-phrased finding rate	—	—	15.5%
<i>Critics evaluated by the benchmark, judging raw frames</i>			
Gemini 3.0 Pro	11.3%	5.5%	16.9%
Qwen3-VL-32B	3.2%	1.0%	14.4%
Qwen3-VL-8B	2.7%	0.8%	12.7%
Claude 4.5 Opus	5.7%	2.3%	10.7%
GPT-5.2	4.0%	3.7%	10.5%
Gemini 2.5 Pro	7.1%	1.1%	10.4%
Gemini 2.5 Flash	4.4%	1.7%	8.2%
Gemini 2.5 Flash Lite	2.0%	0.4%	4.5%
Cosmos Reason 1	0.1%	0.5%	2.8%
Cosmos Reason 2	0.9%	0.5%	1.2%

Table 11 Interim Physion-Eval [80] benchmark table. This work’s rows are measured on the first 601 release clips of a fixed randomized stream (543 with a reported glitch, 58 without): the evidence-conditioned and anomaly-phrased rows report the share of all 601 clips flagged, and the matched-report row the share of the 543 reported-glitch clips whose report a finding matches; the three measures are defined in the text. The benchmark’s rows are transcribed from its Figure 4 and are measured on its full evaluation set: J is its Youden statistic averaged over the five generators, and the flag rate is the share of generated clips called physically unrealistic, pooled over the two viewpoints. A dash in the J columns marks a statistic defined against matched real-world reference videos that the public release withholds, so it cannot be computed for any evaluator run on the release.

6 Framework: Learning as State Optimization

The preceding sections described three artifacts: a physics-guided generator (Section 2), a 1,500-clip human-authored defect benchmark (Section 3), and a physically grounded, tool-using critic whose plan-based adjudication we measure in Section 5. This section states the abstraction that ties them together and the form the framework is built toward. Our thesis is that a physical-reasoning agent should keep everything it has learned in one *human-readable, typed, provenance-carrying state*, and should improve by *optimizing that state*. Every improvement is then an edit to an object a person can read, version, and veto, and the arc of the framework culminates in a system that evolves itself without ever touching a weight.

Sections 2, 3 and 5 report experiments; the remainder of this section sets out the framework and the roadmap. Because all learning is state optimization on a readable S , that boundary falls inside the formalism itself, and Table 13 marks it.

6.1 From ReAct and agent memory to a physical agent with typed state

Agentic reasoning in the ReAct lineage interleaves *reason*, *act*, and *observe* steps, letting a model call external tools and condition on their returns [18, 60, 77]. Program-of-thought variants push this further: the chain of thought is emitted *as a program* that composes tool calls [20, 30, 55, 59, 63, 70]. Orthogonally, a fast-growing agent-memory literature asks where an agent’s accumulated experience should live—episodic transcripts, distilled semantic priors, or reusable procedures [6, 27, 31, 35, 42, 50, 71, 73, 81, 82]. PhyReAct combines these two lines and changes the *carrier* of memory: where these agents store free-text notes or latent vectors, it stores structured physical-verification traces with full provenance. Auditability follows from that choice, and it is the throughline of the framework below.

6.2 The unifying abstraction

PhyReAct maintains a single, human-readable *agent state*

$$S = (\Pi, M, K), \quad (20)$$

and it improves by optimizing S . The reasoning model and the low-level physical experts are frozen, so what the system accumulates is held in S , which is inspectable, diffable, versionable, and revertible. “Training” and “inference” are one readable procedure applied in two directions: one map reads S to act, and channels write to S (or to the generator) to improve.

- Π — **operator library (procedural memory)**. A set of reusable, already-compiled measurement operators. Each operator $o \in \Pi$ is a closure mapping a video and typed arguments to a physical measurement, and it carries an explicit *cost node* $c(o)$ recording its spending (compute, expert calls, latency). Π grows monotonically as recurring reasoning sub-procedures are packaged into named operators.
- M — **experience library (episodic + semantic memory)**. Versioned, human-readable priors distilled from past traces: nearest-neighbor *plans* for similar prompts and *physics-failure priors* over common defect modes. M keeps *evidence* (what was measured) separate from *belief* (what we now expect): evidence is retained as recorded, while belief is defeasible and is overridden by fresh measurement.
- K — **external knowledge channel**. Retrieved external or social knowledge (physical facts, domain conventions, human corrections) admitted only through a candidate-and-pruning discipline. K is the deliberate *open* port that keeps the internal improvement loop from closing on itself (Section 6.6).

The rest of the framework consists of two kinds of maps over this state: a single-episode *read* operator that consumes S and emits an auditable trace (Section 6.3), and *write* channels that update the generator and S from that trace. The six design pillars of the paper are recovered as special cases of read-vs-write over S : P1/P2 and the read side of P4 are “read S to produce a trace”; P3 and P6 are “write S ”; P5 is “write the generator”; and the supervisory side of P4 wraps the writes.

6.3 The single-episode operator (read path): physicalized ReAct as a program over Π

Given a prompt–video pair (p, V) , one episode is a deterministic pipeline that only *reads* S :

$$\text{retrieve: } (\pi_p, m_p, k_p) = r(S, p), \quad (21)$$

$$\text{compile: } (C, G) = \text{Reason}(p, V \mid \pi_p, m_p, k_p), \quad (22)$$

$$\text{act–observe: } R = \text{Exec}_\Pi(G, V), \quad (23)$$

$$\text{adjudicate: } (Y, F) = \Phi(C, R), \quad (24)$$

$$\text{emit: } T = (C, G, R, Y, F). \quad (25)$$

Here C is a set of *typed physical obligations*—explicit, machine-checkable statements of what the video must satisfy to honor the prompt (object presence and count, contact and support, trajectory continuity, depth ordering, on-screen text, audio events). G is a *program over* Π : reasoning takes the form of a composition of operator calls that the model writes and the runtime executes, in the tradition of program-of-thought and modular visual reasoning [20, 30, 63, 70]. Executing G drives the classic reason→act→observe loop of ReAct-style agents [77], but every **act** is a call into a *frozen* physical expert operator—promptable segmentation and tracking [7], counting over its instance identities, the typed physical measurements taken over those tracks, monocular depth, OCR, and audio event detection [21]—registered in Π . This is the sense in which PhyReAct specializes ReAct to physical measurement: it inherits the control-flow harness from mainstream tool-using agents [55, 59] and specializes it with explicit physical obligations and physical-measurement operators (**P1**). This general form is instantiated concretely by the critic of Section 4; Equation (26) below is the same three-valued verdict used there.

Each operator returns a measurement together with a *gate* on its own scope: when an operator’s preconditions are not met (out-of-scope object, untrackable occlusion, no audio track), it emits **abstain**. The adjudicator Φ in (24) is a *deterministic* roll-up from evidence R to a three-valued verdict

$$Y \in \{ \text{supported}, \text{contradicted}, \text{unknown} \}, \quad (26)$$

plus a *contradiction packet* F that localizes, for each unmet obligation, which measurement contradicts it and where. **unknown**, and **abstain** for out-of-scope obligations, are first-class outcomes of the codomain, so declining to certify is available to the critic as a verdict in its own right. The emitted trace $T = (C, G, R, Y, F)$ carries full provenance—every element of Y is back-traceable to the operator calls in G and the measurements in R .

CoT-as-program and cost nodes (P2). Because G is a program, recurring sub-procedures (e.g. “verify a bounce = track the object, detect the contact frame, check the pre/post vertical-velocity sign flip”) can be packaged, named, and re-registered as new operators in Π . Each operator carries its cost node $c(o)$, so a program has a well-defined budget $c(G) = \sum_{o \in G} c(o)$ and the reasoner can trade breadth of measurement against spending. Over many episodes Π scales up into a growing procedural memory, and reasoning increasingly becomes *composition of trusted operators* rather than re-derivation from scratch. What is instantiated today is a fixed operator set Π_0 and a fixed compiler; the *scaling-up* of Π via distillation is roadmap (Section 6.9).

6.4 The outward channel: verification-guided refinement (P5)

The read path produces, for a failed clip, a localized contradiction packet F (defined in Section 4.9). The critic the outward *control channel* maps F into a *local edit of the generator’s control signal* and regenerates:

$$V' = \text{Gen}(p, \delta(F)), \quad \delta : F \mapsto \Delta(\text{control}), \quad (27)$$

where Gen is the physics-guided pipeline of Section 2 (MuJoCo simulation [64] driving a Wan 2.2 video model [65] through VACE control conditioning [29]), and $\delta(F)$ is an edit to the simulation and control inputs addressing the violated obligation. Since the control signal is a rendered field over space and time, a contradiction localized in both maps onto a local edit of the conditioning stream. The critic here is independent and frozen, so the grading signal does not originate in the generator being graded, unlike self-evaluating refinement loops [66] and verification-guided repair [11]. Both endpoints of (27) are built (Sections 2 and 5); the map δ and the closed loop are roadmap.

6.5 Auditability of the state and the trace

Auditability follows from the formalism. The learned object is the readable state S and every conclusion is a deterministic roll-up Φ over provenance-carrying measurements, so any verdict Y replays back to the operator calls in G and the measurements in R , and any state change $S \rightarrow S'$ is a human-readable diff of Π , M , or K . Weight-free self-evolution and its human supervision both rest on that property: the reviewer inspects readable objects rather than gradients.

Auditability determines what a verdict can be taken to mean. A **supported** verdict is a critic judgment, where the ground truth in this paper is human annotation. Determinism in the roll-up leaves evaluator drift in the underlying operators and the compiler untouched. A prompt surviving the checks is *plausible* with respect to the obligations tested, and a static critic improving says nothing about the generator.

6.6 Weight-free context-training self-evolution

The framework’s inward channel provides: a critic that improves itself without updating a weight. Everything above—a readable state, a program over operators, provenance traces, an independent verdict—exists so that the system’s own experience can be written into S under a guard. This inward *context-training* channel (P3) is the last of the framework’s components to be instantiated.

Given a trace T and feedback ϕ (human labels, downstream outcomes, or a later, more complete measurement), we distill *candidate* state edits and accept them under a strict guard:

$$(\Delta\Pi, \Delta M, \Delta K) = \text{Distill}(T, \phi), \quad (28)$$

$$S' = \text{Accept}(S, \Delta\Pi, \Delta M, \Delta K). \quad (29)$$

Both successful and failed episodes are distilled (hindsight over traces [31, 50]): a success may add a plan to M or an operator to Π ; a failure may add a physics-failure prior or a scope restriction. This

form of self-evolution leaves weights unchanged, in the spirit of context-distillation and experience-memory agents [6, 27, 50, 60, 71, 81, 82], and it treats deployment-time adaptation as *learning as state optimization*. Acceptance is governed by

$$S' = \begin{cases} S \oplus \Delta, & \text{if } U_{\text{val}}(S \oplus \Delta) > U_{\text{val}}(S), \\ S, & \text{otherwise (do-nothing elitism),} \end{cases} \quad (30)$$

where U_{val} is a *held-out proxy* score and Δ ranges over distilled candidates. Candidates that do not strictly improve the proxy are pruned, and when none improves it the state is left unchanged. That default makes the proxy monotone across updates, bounding how far a distilled candidate can degrade S .

Admitting external knowledge (P6). A system whose only inputs are its own traces—CoT $\rightarrow$ feedback $\rightarrow$ context $\rightarrow$ CoT—is a *closed* loop that can amplify its own biases: distilled priors condition future reasoning, whose traces distill into more priors. The external knowledge channel K is the opening provided for this. **PhyReAct** may issue retrieval or ask for external or social knowledge (physical facts, human corrections, domain conventions) and admit it into K under the same candidate-and-pruning discipline as (30): a retrieved candidate is admitted only when it improves the held-out proxy, and K -sourced facts carry a provenance tag distinct from distilled beliefs, so an admission can be reverted. External knowledge therefore enters under the same elitism that governs S .

Every prior distilled into M is *critic-derived*: a summary of what **PhyReAct**’s own verification loop concluded rather than verified truth. The external channel K is roadmap. The inward channel is instantiated, and [section 6.7](#) measures it.

6.7 Distilling lessons from recorded misses

We run (28) over episodes in which a plan missed a flaw a human annotator recorded. Distill reads the trace and the miss and writes a lesson naming a property of the claim and the action it licenses — *when a prompt specifies a singular entity or exact quantity, create a separate cardinality check and verify that count across the beginning, middle and end* — and a curation pass groups the candidates into 33 *families* of equivalent advice and keeps one lesson from each. Those 33 lessons are appended to the planner’s rubric; a family is the group a kept lesson stands for, and its name is what the analysis below uses to say which kind of defect a lesson is about. Only M changes: the operator library Π and the external channel K are untouched, so $S' = (\Pi, M', K)$.

Two aspects of the implementation depart from the formalism above. All 33 lessons are supplied on every clip. We built the selective form as well — one relevance call per lesson–claim pair — and it is the more expensive of the two. Choosing among the 33 lessons costs 132 model calls for a four-claim prompt and 495 for a fifteen-claim one, whereas supplying all of them costs no model calls and adds 9 kB to a planner prompt already 17.8 kB long.

The acceptance gate (30) is also not applied. U_{val} is defined on rules that have appeared in a run, so it cannot score a rule written from a fresh miss. Applied as a diagnostic to 80 earlier candidate rules over a ten-clip decision set, it admitted none: 39 did not meet its minimum-evidence criterion, and 41 never triggered on a clip in that set. The intervention evaluated below is therefore the curated set of 33 lessons, without the gate.

Protocol. Lessons are distilled from a pool of experience clips and tested on clips held out from it. Excluding every clip that shares an identifier with a clip in the experience pool leaves 363 candidates; we then exclude the 57 whose prompt restates the prompt of an experience clip, so that no lesson is evaluated on the prompt it was written from. The evaluation set is the remaining 306 clips and the 502 flaws they carry, and every condition is scored on those same flaws. The untaught condition loads an empty lesson file through the same retrieval path. All are scored by the matching protocol of [section 5.1](#), under which a flaw counts as *found* when the critic’s findings cover it wholly or in part, the same criterion as [table 7](#).

Varying how much experience the lessons are distilled from turns the comparison into a curve. Five conditions are run: no lessons, then 8, 13, 19 and 33 lessons distilled from 25, 50, 75 and 117 experience clips. Each pool is a prefix of the next, so the lesson sets are nested — every set is the previous one plus the families

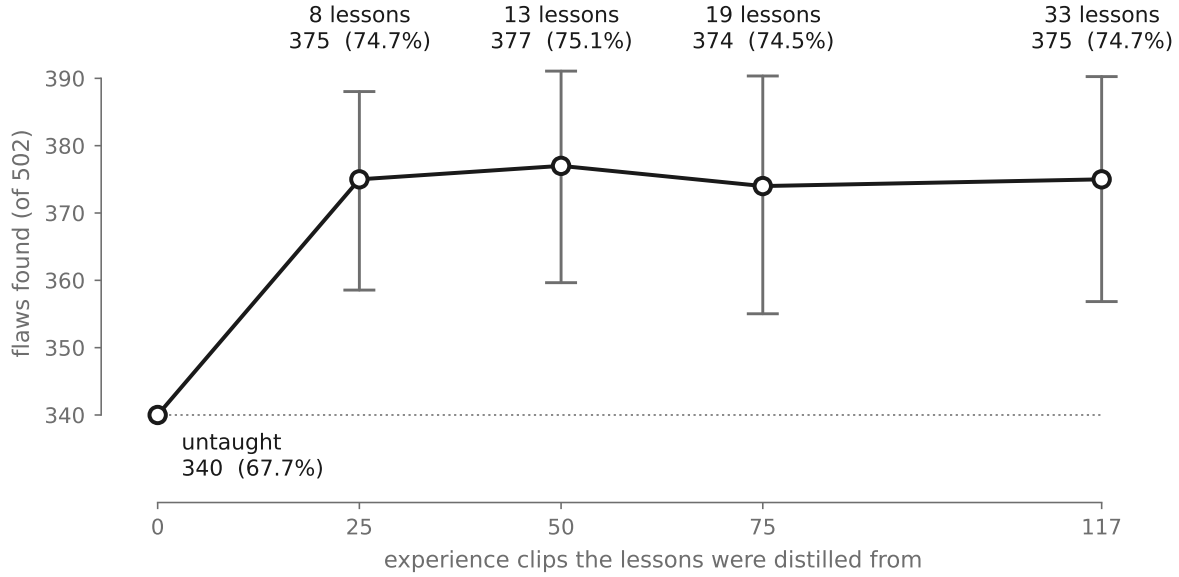


Figure 17 Flaws found against the number of experience clips the lessons were distilled from, over the same 502 held-out flaws in every condition. The dotted line marks the untaught condition, 340 of 502 (67.7%). Whiskers are 95% intervals on the change from untaught, in flaws, computed from the flaws that switch status, since the conditions score the same flaws. Every taught condition exceeds the untaught one, while pairwise comparisons among the taught conditions return $p \geq 0.8$.

the additional experience produced, with none revised away — and the conditions differ only in how much experience was available to distil from.

Result. With all 33 lessons the planner finds **375** of the 502 flaws (**74.7%**) against 340 untaught (67.7%). Relative to the untaught condition, 62 flaws switch from missed to found and 27 switch the other way, a net +35 at $p = 3 \times 10^{-4}$ under a paired sign test.

Distilling from more experience does not extend that gain. The four taught conditions find 375, 377, 374 and 375 flaws (figure 17); each exceeds the untaught condition by +34 to +37 at $p < 10^{-3}$, and no two of them differ by more than 3 flaws, with every pairwise sign test returning $p \geq 0.8$. Comparing the smallest and largest lesson sets directly, 36 flaws move each way for a net of zero. The gain is already present with lessons distilled from 25 clips and does not increase across the larger experience pools. Before the prompt filter, on 363 clips and 626 flaws, the five conditions find 428, 463, 470, 473 and 465: every taught condition again exceeds the untaught one, and the taught conditions again do not separate from each other, spanning 10 flaws with $p \geq 0.3$ for every pairwise comparison among them.

What the additional experience changes. Although the later lessons do not change the total number of flaws found, they change which flaws are found. Comparing the 33-lesson condition against the 8-lesson one directly, 36 flaws move each way. Of the 36 it newly finds, 26 were missed in the untaught condition too and 10 were found untaught but missed at 8 lessons; of the 36 it no longer finds, 19 were untaught misses the smaller set had recovered and 17 were found in both the untaught and 8-lesson conditions. Distilling from a further 92 clips therefore changes which individual flaws are found while leaving the total unchanged.

Effect by flaw category. Because the conditions score the same flaws, the informative quantity is the paired change, and its interval comes from the flaws that switched status (table 12). Taking the untaught and 33-lesson conditions, two categories have a 95% interval excluding zero. *Action or event occurs*, the largest category at 211 flaws, gains 17 of them, a rise of 8.1 percentage points with a 95% interval of [+2, +12] points;

Flaw category	Flaws	Flaws found, with ℓ lessons					Change	95% interval (pts)	Families
		$\ell=0$	8	13	19	33			
action or event occurs	211	146	169	169	166	163	+17 (+8)	[+2, +12]	20
on-screen text	73	53	63	56	62	61	+8 (+11)	[+2, +15]	1
attribute	46	29	30	32	31	33	+4 (+9)	[-2, +12]	2
order and timing	22	11	13	13	13	14	+3 (+14)	[-9, +27]	3
how many	19	9	9	11	11	12	+3 (+16)	[-10, +31]	2
object identity	52	39	41	39	42	41	+2 (+4)	[-3, +7]	4
camera and style	14	11	11	12	11	12	+1 (+7)	[-4, +7]	0
spatial relation	45	28	29	30	28	28	+0 (+0)	[-10, +10]	1
physical motion	15	9	8	12	9	9	+0 (+0)	[-11, +11]	0
All flaws	502	340	375	377	374	375	+35 (+7)	[+3, +10]	33

Table 12 Effect of the distilled lessons on each flaw category, over the same 502 held-out flaws in every condition. A flaw is *found* when the critic’s findings cover it wholly or in part, as in [table 7](#). The five middle columns count that category’s flaws found with no lessons and with 8, 13, 19 and 33 lessons, distilled from 25, 50, 75 and 117 experience clips; **Change** is the untaught-to-33 difference in flaws, with percentage points in brackets, since several categories carry few enough flaws that a rate alone would overstate the movement. Because the conditions score the same flaws the comparison is paired: the interval on the change is computed from the flaws whose found status switched, and is given in percentage points of that category’s flaws. **Bold** marks the rows whose interval excludes zero. **Families** counts, over all 33 lessons, those whose family name refers to that category, matching on the words in the name. The other category, containing 5 flaws in all, is omitted from the category rows; the *All flaws* row includes those flaws.

on-screen text gains 8 of 73, a rise of 11.0 points with [+2, +15]. Every other interval contains zero. Reading the smaller movements as rates overstates them: the +7 points for camera and style are one flaw of 14, and the +16 for counting three of 19. The number of flaws found rises in both of the severity grades that carry the bulk of the set, by 19 of 288 at severity 3 ($p = 0.01$) and by 13 of 163 at severity 4 ($p = 0.03$); the remaining 51 flaws sit at grades 1, 2 and 5.

Where the lessons point. All eight families induced from the first 25 clips concern actions and the roles within them: decomposing an action into its parts, verifying that the asserted action occurred, checking the manner of a motion, separating the participants of a transfer, attributing a subject correctly, preserving a reference frame, reconciling checks that disagree, and providing a fallback when a measurement fails. None mentions text, counting or spatial relations. In the 8-lesson condition *action or event occurs* gains 23 flaws and *on-screen text* 10, together 33 of that condition’s +35.

[Table 12](#) assigns a family to a flaw category when the family’s name contains a word that names it, and to *action or event occurs* otherwise; on that rule seven of these eight names go to action or event occurs and the eighth, *subject attribution error*, to object identity. The later 25 families name their targets more narrowly: 13 again go to action or event occurs — giving the 20 the table records — three to order and timing, three to object identity, two to counting, two to attributes, one to spatial relations and one to on-screen text. Between the 8- and 33-lesson conditions the six non-action categories the later families name move by +1 flaw for order and timing, 0 for object identity, +3 for counting, +3 for attributes, −1 for spatial relations and −2 for on-screen text: a net of +4. Action or event occurs, which draws 13 of the 25, falls 6.

Category gains are therefore not consistently aligned with the lesson names. Nothing among the first eight families mentions text, yet on-screen text gains 10 of 73 flaws at that condition, the largest gain of any category in percentage points; and the later families, which do name six categories between them, move those categories by +4 flaws in total. On-screen text is the clearest example in both directions: it gains ten flaws before any lesson refers to text, and loses two after one does. This pattern is consistent with the first lessons supplying a general instruction to separate a compound requirement into checkable parts, one that applies to prompts well outside the categories the lessons were written from. The plan measurements below assess that interpretation.

Pillar	Where it lives in the loop	Status
P1 Physicalized ReAct + operators	compile / act-observe over frozen experts [7, 21]	Phase 0 (built)
P2 CoT-as-program + cost nodes	program G over Π , budget $c(G)$	Phase 0 partial
P4 Auditable trace + three-valued Y	trace $T = (C, G, R, Y, F)$, provenance	Phase 0 (artifacts)
P5 Verification-guided refinement	outward control channel δ , Gen	roadmap
P2 ⁺ Operator-library scale-up	procedural-memory growth in Π	roadmap
P6 Breaking the loop	external channel K , candidate+prune	roadmap
P3 Context-training self-evolution	inward channel Distill (28)	Phase 0 (measured, gate not applied)
P3 ⁺ Acceptance gate	elitism (30) over distilled candidates	roadmap
P4 ⁺ Human oversight	supervision + deployment gate	roadmap

Table 13 Each pillar is one component of the same loop. Only Phase 0 rows are backed by measurements; solid arrows in Figure 18 correspond to them, dashed arrows to the roadmap rows, which are unmeasured.

Effect of the lessons on the plan. A lesson names a property of a claim and the check it licenses, which suggests the lessons split one claim into several checks. That is the smaller of the two movements they produce. Between the untaught and 33-lesson conditions, checks per claim rise from 1.13 to 1.18, while the number of typed physical claims the planner compiles from the prompt rises from 2,905 to 5,106 over the 306 evaluation clips, or 9.5 to 16.7 per clip. Each lesson is a conditional instruction of the form *when a prompt specifies X* , and the change is concentrated in how many claims the prompt is compiled into rather than in how many checks each claim receives, which is the pattern the decomposition reading predicts.

The same measurement locates where the curve flattens. Claims per clip continue to rise across all five conditions — 9.5, 15.1, 16.0, 16.1 and 16.7 — so the later lessons go on changing the plan, but the number of flaws found does not rise with them. On-screen text shows the pattern at clip level. Of the 56 evaluation clips carrying a text flaw, the number receiving at least one text check does not rise: 41 untaught against 40 with 33 lessons. The two sets overlap on 37 clips, with three checked only when taught and four only when untaught. The number of text checks written across those clips rises from 81 to 136, and the 8-lesson condition already produces 114 of them. The lessons therefore add checks without widening the set of clips on which text is checked at all, and 25 experience clips supply 33 of the 55 added.

6.8 Human oversight

A human is kept in the loop at a few specific gates: a reviewer may audit a trace T , veto or edit a proposed state update in (29), and approve whether a refined generation (27) or a new state S' is deployed. Human feedback also enters Distill as a first-class signal ϕ . Because updates are readable diffs of S , oversight consists of reading and approving them. The reviewing tooling and deployment protocol are roadmap; the artifacts they act on — traces, three-valued state, provenance — exist today.

6.9 Built and roadmap components

Because all learning is state optimization on a readable S , we can mark built-vs-roadmap directly on the formalism, and each pillar is one component of the same loop (Table 13). **Instantiated today (Phase 0):** the frozen generator pipeline Gen (Section 2); the 1,500-clip defect benchmark that exercises the read path (Section 3); the episode read operator (21)–(25) with a fixed operator set Π_0 , typed obligations C , three-valued deterministic adjudication Φ , and provenance traces T ; the flaw-level measurement in Section 5 against human labels (228 of 304 for PhyReAct, 164 for question decomposition, recall-only, on the development core, which is not held out); and the inward distillation of M via (28), measured on a held-out split in section 6.7 (375 of 502 against 340 untaught). **Roadmap (Phase 1+):** the outward control map δ and the closed critic→generator loop (27); the scaling-up of Π ; the acceptance gate (30) that admits distilled candidates on a measured proxy improvement; the external knowledge channel K ; and human-in-the-loop deployment at scale. Every roadmap item is a *write* path in the same formalism whose *read* counterpart is already exercised. The executable plan follows in Section 7.

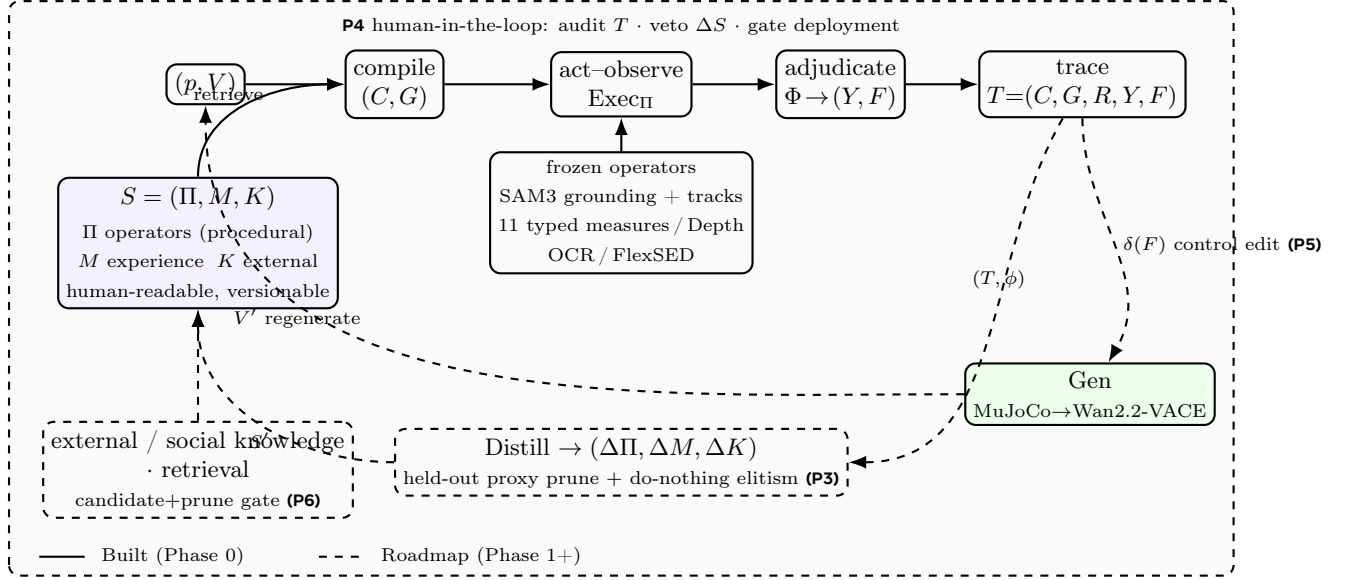


Figure 18 State-optimization view of **PhyReAct**. Everything the system learns lives in one human-readable state $S = (\Pi, M, K)$. A single *read* operator (top, solid = built) conditions on S , compiles (p, V) into typed obligations C and a program G over Π , runs frozen physical operators, and emits a provenance-carrying three-valued trace T . Two structurally dual *write* channels (dashed = roadmap), both driven by the same independent frozen critic, close the loop: the outward control channel edits the generator’s control signal via $\delta(F)$ and regenerates (P5); the inward context-training channel distills T into candidate state edits accepted only under held-out proxy pruning with do-nothing elitism (P3). The external channel K breaks the internal loop (P6). A human-in-the-loop layer wraps both channels (P4). Solid arrows are instantiated in [Sections 2, 3](#) and [5](#); dashed arrows are the roadmap of [Section 7](#).

7 An Executable Roadmap

The framework of [Section 6](#) defines one learned object, the state $S = (\Pi, M, K)$, and three maps over it: a read operator (built) and two write channels (roadmap). We phase the work accordingly. Phase 0 is the read path and the two frozen endpoints already instantiated in this paper; Phase 1 moves the evaluation onto a held-out split scored for precision as well as recall; Phases 2–6 exercise the write paths, each with a measurable success metric and each a strict superset of what precedes it. No number below the Phase 0 line is a result; Phases 1+ state *targets and protocols*, not outcomes.

Phase 0 — Built: the read path and frozen endpoints (done).

- **Delivered.** The physics-guided generator Gen (MuJoCo $\rightarrow$ Wan2.2-VACE, [Section 2](#)); the 1500-clip human-authored defect benchmark ([Section 3](#)); the episode read operator with fixed Π_0 , typed obligations C , three-valued deterministic adjudication Φ , and provenance traces T ([Section 4](#)); and the flaw-level measurement against human labels (228 of 304 for **PhyReAct** against 164 for question decomposition, on the development core, non-held-out, [Section 5](#)).
- **Covers.** P1 (physicalized ReAct read path), and the read side of P2 (programs over Π_0) and P4 (provenance traces).
- **Honest limits.** Recall-only, single non-held-out split; critic verdicts are judgments scored against human labels, not ground truth; a static evaluator result is not evidence any generator improved; the two write channels are not yet exercised.

Phase 1 — Held-out, precision+recall evaluation with provenance completeness.

- **Goal.** Promote the critic result from dev-core / recall-only to a held-out, precision+recall protocol, and guarantee every verdict carries a complete provenance trace T .

- **Needs.** A frozen held-out split of the benchmark disjoint from dev-core; a provenance schema for (C, G, R, Y, F) .
- **Experiments.** Plan-based vs. per-claim adjudication on held-out clips with full confusion matrices; audit that every **contradicted** verdict localizes to frame-level evidence via F .
- **Success metric.** Held-out plan-based accuracy $\geq$ per-claim with non-degrading precision; complete provenance coverage.
- **Risk.** Held-out gap reveals dev-core overfitting; the three-valued codomain inflates **unknown** to dodge hard cases — mitigate by reporting the **unknown** rate alongside accuracy.

Phase 2 — Operator library with cost accounting (P2).

- **Goal.** Turn fixed Π_0 into a growing, versioned Π whose operators carry cost nodes $c(o)$ and whose programs have measurable budgets $c(G)$.
- **Needs.** An operator registry with provenance and versioning; a compiler that detects recurring G -sub-programs and promotes them into named operators; per-operator cost instrumentation wired into Φ .
- **Experiments.** Track the coverage/cost frontier as $|\Pi|$ grows; ablate frozen Π_0 vs. promoted Π at matched budget.
- **Success metric.** A Pareto improvement in obligation coverage per unit cost over the Phase 0 fixed pipeline at equal or lower $c(G)$.
- **Risk.** Library bloat and stale operators; promoted operators encode clip-specific shortcuts — mitigate with usage-frequency thresholds and held-out re-validation before promotion.

Phase 3 — Context-training self-evolution of M (P3).

- **Goal.** Distill traces (success and failure) into human-readable, versionable priors in M — nearest-neighbor plans and physics-failure priors — accepted under held-out proxy pruning with do-nothing elitism (30), keeping evidence separate from belief.
- **Needs.** A distillation function Distill ; a held-out validation split U_{val} (new data, disjoint from the Phase 0/1 core); a diffable memory store.
- **Experiments.** Compare {no memory, retrieval-only, distilled M with elitism, distilled M without elitism} on the held-out split; a pollution stress test injecting wrong priors and checking elitism rejects them.
- **Success metric.** Monotone non-decreasing held-out proxy across accepted updates, a positive gap of “ M with elitism” over both baselines, and zero acceptance of injected harmful priors.
- **Risk.** Belief leaking into evidence, or proxy overfitting; mitigate with the evidence/belief split and periodic held-out refresh.

Phase 4 — Closing the control channel critic $\rightarrow$ generator (P5).

- **Goal.** Implement $\delta : F \mapsto \Delta(\text{control})$ of (27), converting a contradiction packet into a *targeted* local edit of the MuJoCo/VACE control signal, and regenerate.
- **Needs.** A parameterization of Wan2.2-VACE control inputs by obligation type; a differentiable-or-heuristic edit policy δ ; a re-adjudication harness.
- **Experiments.** On clips with **contradicted** verdicts, compare {blind resampling, prompt-only edit, targeted $\delta(F)$ edit}; report the fraction of previously violated obligations that flip to **supported** after regeneration, *human-verified*, and re-run the full obligation set C post-edit to report net defect change.
- **Success metric.** Targeted $\delta(F)$ regeneration repairs a significantly higher fraction of localized violations than blind resampling at equal generation budget, confirmed by human annotation (not by the critic alone), with no rise in unrelated-obligation contradictions.
- **Risk.** Goodharting the critic or whack-a-mole edits that fix one obligation while breaking another; mitigated by human verification and the net-defect audit.

Phase 5 — Breaking the loop with external knowledge K (P6).

- **Goal.** Open the internal CoT→feedback→context loop by admitting external/social knowledge and retrieval into K under the same candidate-and-pruning discipline, with a do-nothing fallback.
- **Needs.** A retrieval/ingestion interface; a candidate gate scored by U_{val} ; provenance tagging that marks K -sourced facts distinctly from distilled beliefs.
- **Experiments.** On a bias-probe subset (defect modes under-represented in the internal traces), compare closed-loop ($K = \emptyset$) vs. open-loop (K admitted with gating); a noise-injection test on retrieved sources.
- **Success metric.** Reduced error on the bias-probe subset with no regression on the general held-out split; zero admissions when no candidate improves the proxy (fallback correctness).
- **Risk.** Context pollution from noisy or off-distribution sources; mitigated by the same elitism gate as S , plus source provenance so bad admissions are revertible.

Phase 6 — Human-in-the-loop deployment at scale (P4).

- **Goal.** Operate both channels under human supervision: reviewers audit traces T , veto/edit candidate state updates, and gate deployment of refined generations and new states S' .
- **Needs.** Trace-replay and state-diff review tooling; an approval workflow feeding Distill and K ; audit logs over $S \rightarrow S'$ with rollback.
- **Experiments.** Held-out study measuring reviewer time per audited trace, inter-annotator agreement on state-update acceptance and on three-valued verdicts, and the catch-rate of injected bad updates that elitism alone would have missed.
- **Success metric.** Auditing cost per trace low enough to be practical, with human review reliably catching a measurable residual of harmful updates beyond the automatic gate.
- **Risk.** Reviewer fatigue and rubber-stamping (which would reintroduce an unaudited closed loop); mitigate by surfacing only elitism-borderline updates for human attention.

Ordering rationale. Phase 1 secures the evaluation; Phases 2–3 grow the two components of S that the inward channel writes (Π , M); Phase 4 exercises the outward channel; Phase 5 opens K ; Phase 6 wraps everything in human oversight. Each phase reuses the Phase 0 read path unchanged, so at every step the system remains weight-free and auditable by construction.

8 Related Work

8.1 Physics-aware video generation and refinement

Controllable diffusion separates appearance synthesis from structural guidance. ControlNet introduced trainable branches for conditions such as depth and segmentation [79], VideoComposer extended compositional control to space and time [67], and Wan and VACE provide a scalable video backbone and a unified interface for reference, mask, and structural conditions [29, 65]. These methods determine how a condition enters a generator, but not whether that condition or the rendered pixels obey physics.

A complementary line derives guidance from physical reasoning or simulation. MotionCraft warps diffusion latents with simulated optical flow, while PhysGen estimates scene properties, performs rigid-body simulation, and refines the result with a video prior [38, 47]. DiffPhy injects language-model-inferred physical context through supervised fine-tuning [78]; PhyT2V iteratively rewrites prompts from physical-rule and mismatch reasoning [72]; and VLIPP turns vision-language reasoning into a coarse motion plan [75]. PhyCo learns continuous controls for physical properties from simulation data [49]. More tightly coupled systems place a simulator inside generation: PSIVG reconstructs a generated template, simulates object trajectories, and feeds them back at test time [16], while MoReGen coordinates language agents, simulation, and rendering for code-based synthesis [2].

Evaluation signals have also entered the generation loop. Binary VLM feedback has been used for preference optimization of object interactions [17]; Video-T1 searches candidate denoising trajectories with test-time verifiers [37]; and VideoRepair converts detected semantic misalignment into localized regeneration [32]. Our

generation testbed is deliberately narrower: known MuJoCo state is rendered into metric depth or silhouettes for a frozen Wan 2.2-VACE model. This avoids inverse reconstruction and generator training, but a physically coherent control signal does not certify that the synthesized pixels preserve its physics. The present work therefore evaluates the generator and critic separately; using **PhyReAct** evidence to guide generation remains future work.

8.2 Generated-video evaluation and physical benchmarks

General-purpose suites measure complementary aspects of generated video. VBench, T2V-CompBench, FETV, and EvalCrafter aggregate perceptual quality, temporal consistency, and prompt-alignment criteria [28, 39, 40, 62], while VideoScore and VideoScore2 learn to predict fine-grained human judgments [22, 23]. Prompt decomposition is also established: TIFA and the Davidsonian Scene Graph turn text into questions or propositions [9, 26], and VQAScore evaluates a text condition with an image-to-text model [36]. These approaches are valuable for ranking systems, but their final scores do not by themselves expose a measurement trace for a specific physical allegation.

Physics-specific evaluation has rapidly broadened. VideoPhy and VideoPhy-2 assess semantic adherence and physical commonsense, with VideoPhy-2 additionally labeling candidate physical rules as followed, violated, or undetermined [3, 4]. PhyGenBench spans common physical phenomena [44], Physics-IQ tests whether generative models encode physical principles [48], and PhyCoBench compares observed motion with optical-flow-guided future-frame prediction [8]. PhyWorldBench extends coverage to fundamental, composite, and deliberately anti-physical prompts, with prompt-specific evaluation standards [19]. These benchmarks provide essential generator-level comparisons and increasingly structured criteria. Our question is complementary: given an arbitrary prompt and clip, can an evaluator expose which physical obligation failed, what instrument measured it, when the supporting evidence occurred, and why insufficient evidence produced an abstention rather than a pass?

8.3 Agentic tool use and neuro-symbolic video reasoning

ReAct established the reasoning–action–observation loop, in which tool results can change the next reasoning step [77]. Related work explores how such behavior is acquired and organized. Toolformer learns when and how to insert API calls through self-supervision, while GPT4Tools transfers multimodal tool use to open language models through self-instruction [55, 74]. HuggingGPT uses an LLM controller to plan and route subtasks across expert models [59], while ControlLLM and ToolChain* search tool-call graphs or action trees [41, 83]. ReWOO and LLMCompiler instead plan before execution and make tool dependencies explicit [30, 70]. Feedback-based agents close another loop: Reflexion stores linguistic feedback in episodic memory across trials, whereas CRITIC invokes external tools to validate and iteratively revise an answer [18, 60].

Multimodal systems specialize these ideas to perception. Visual programs such as VisProg, ViperGPT, and CodeVQA compose vision modules through generated programs [20, 61, 63], while ProViQ, MoReVQA, the VideoAgent systems, and VideoTree adapt modular or selective observation to video [12, 15, 46, 68, 69]. Recent video agents make observation itself adaptive: CAViAR invokes modules and uses a critic to select reasoning traces, and LensWalk controls temporal scope and sampling density [33, 45]. Their primary setting is video question answering rather than physical verification of generated content.

Three systems are particularly close. VideoGen-Eval combines LLM prompt structuring, MLLM judgments, and temporal patch tools for multidimensional generated-video evaluation [76]. NeuS-V translates prompts into temporal-logic specifications, maps frame-level proposition confidence to a video automaton, and uses probabilistic model checking for alignment [58]. NeuS-E then identifies a weak proposition and failure frame and edits or regenerates the affected suffix [11]. Thus, agentic evaluation, prompt-to-temporal specification, and verification-guided repair all predate this work.

PhyReAct is a grounded instance of this ReAct family specialized for physical verification. Rather than let perception decide the verdict in prompt space, the reasoning model compiles a prompt into *typed physical obligations*, and frozen physical operators—counting, tracking, segmentation, depth, OCR, audio, and open-vocabulary semantics—return measurements rather than judgments. Observations then gate and scope the obligations that remain worth measuring, and a fixed set of composition rules rolls the measurements into a

three-valued state (**supported**, **contradicted**, or **unknown**, surfaced as **abstain** when evidence is insufficient) while retaining claim-level provenance. This plan-before-act discipline and its code-as-plan compilation inherit directly from ReWOO, LLMCompiler, and ViperGPT [30, 63, 70]: the control flow follows the plan-execute branch of ReAct, where observations gate and scope already-declared measurements rather than trigger free-form re-planning or the on-the-fly invention of new tool sequences.

Where **PhyReAct** departs from the ReAct lineage is memory: classic ReAct agents accumulate experience within a single episode, whereas **PhyReAct** carries lessons distilled from one set of clips into the planning of a disjoint set (section 6.7). We survey the relevant literature separately in section 8.4.

These positions clarify our relationship to the closest neuro-symbolic evaluators. NeuS-V offers no formal guarantee when its learned perception is wrong [58], which is precisely why **PhyReAct** keeps measurements separate from verdicts and preserves per-claim provenance, so that a wrong measurement remains locally attributable rather than silently absorbed into a score. NeuS-E [11], and more broadly Reflexion [60] and CRITIC [18], produce diagnoses but stop short of turning them into autonomous, online correction of the underlying generator; converting our audited critiques into such a control channel (section 4.9) is a future direction we outline rather than an implemented result.

8.4 Agent memory, context-training, and self-evolving agents

Because **PhyReAct** is designed to evolve by optimizing a human-readable state $S = (\Pi, M, K)$ rather than by updating weights (section 6), it inherits from a fast-growing literature on where an agent’s experience should live and how it should be revised. We organize that literature along the three components of S and, orthogonally, along the storage medium.

Where experience is stored: explicit state versus weights. A central question is whether accumulated experience should be internalized into parameters or externalized into a readable store. Evidence that explicit, retrievable memory can outperform implicit weight-internalization on multi-hop reasoning [82] motivates our commitment to a weight-free state: the reasoning model and physical operators stay frozen, and all learning is an edit to Π , M , or K . Context-training makes this precise by treating deployment-time adaptation as optimization of a textual, human-readable context rather than gradient descent [27], and self-evolving reasoning memories such as ReasoningBank show that curating what an agent remembers—distilled from both successful and failed trajectories—improves it without any parameter update [50, 60]. Surveys of agent memory chart this design space and the accompanying management lifecycle [35, 81].

Procedural memory (Π): reusable operators and programs. The operator library Π is a procedural memory in the sense of PlugMem, which distills raw trajectories into reusable, provenance-carrying procedures indexed for retrieval [73], and of the program-of-thought agents whose chains of thought are executable compositions of tool calls [20, 30, 63, 70]. **PhyReAct** specializes this to physical measurement: each compiled check is a cost-annotated closure that can, in principle, be named and reused across clips.

Episodic and semantic memory (M): distilling and refining priors. The experience library M corresponds to episodic and semantic memory. A-Mem maintains a persistent, self-organizing episodic store of past interactions [71]; ViLoMem distills multimodal experience through grow-and-refine error attribution with no gradient updates [6]; and M3-Agent, the closest analogue in the video domain, builds an entity-centric multimodal memory graph over per-clip observations [42]. We adopt Hindsight’s discipline of separating recorded *evidence* from derived *belief*, so that traceable updates never silently overwrite the raw observations an auditor may later revisit [31].

External knowledge (K) and pollution control. Because a loop that recycles only its own conclusions can amplify its biases, K admits external knowledge, but under a candidate-and-pruning discipline with a do-nothing fallback. This follows the finding that naively equipping an agent with retrieval can degrade it through context pollution, and that maintaining and pruning candidate contexts against a held-out signal is what turns external information into consistent gains [27].

Positioning. What distinguishes PhyReAct within this literature is where memory is held: a structured physical-verification trace with full provenance and a three-valued verdict, where these systems accumulate free-text notes or latent vectors. A trace of that form is auditable as accumulated, which is the property the roadmap of [section 7](#) depends on. None of the cross-clip machinery cited above runs in this report.

8.5 Diagnostic supervision for generated-video failures

Human supervision for generated video ranges from holistic ratings and preferences to dimension-level explanations. VideoGen-Eval evaluates human agreement over more than 12,000 videos generated from 700 structured prompts [76]. Physion-Eval is closer to flaw diagnosis: its 12,718 generated videos and 10,990 expert reasoning traces cover 22 physical categories, temporally localize failures, and use corresponding real-world videos as references [80].

Our 1,500-clip corpus is complementary. Its 2,582 human-written records bind each flaw to the violated prompt fragment and a rationale, severity, and confidence, with an optional time span and box track, which supports open-vocabulary allegation-to-flaw matching without a matched real-world reference. Its all-flawed, single-annotator construction supports recall measurement alone, and yields neither the precision nor the agreement statistics of a balanced, redundantly annotated benchmark.

9 Limitations

Generation. The control signal comes from a simulator, so scene diversity is bounded by what is simulated, and the mapping from simulation to prompt appearance is not itself verified for physical consistency.

Benchmark. The corpus is recall-only (no clean clips), so precision is not directly measurable; flaws are single-annotator with no inter-annotator agreement; the scope/category tags are machine-derived; and the source generator of the clips is unrecorded. Repeatedly designing against one frozen set risks overfitting to it even without training on it.

Critic. The spatial relation predicate and negative-event verification are limited; the typed-plan layer does not yet dispatch the full specialist registry; binding remains a central failure mode, since the compiler enforces symbolic identifier consistency but cannot guarantee that heterogeneous tools localized the same physical instance; and the reasoner-mediated sub-checks reduce but do not remove semantic opacity.

Self-evolution. Four limits bound what [section 6.7](#) establishes. The acceptance gate (30) is not exercised, so the measurement covers distillation without the admission discipline that is meant to keep a bad lesson out; a rule written from a fresh miss has no proxy history, and closing that gap needs a U_{val} defined on unrun rules. The taught planner reads 1.76 times as many claims out of a prompt as the untaught one, and flaw recall rewards asking more, so the effect combines a better-targeted plan with a larger one; a run holding the taught arm to the untaught arm’s question budget would separate them. The lessons are induced once from a fixed pool rather than accumulated online, so the curve of [figure 17](#) measures the effect of the amount of experience distilled, not of accumulating it during deployment; the shape beyond 117 experience clips is unmeasured. Finally, the 502-flaw held-out set is a different population from the 304-flaw evaluation core used everywhere else in this paper, and the two rates should not be compared directly. Every distilled prior is *critic-derived* rather than ground truth, and retrieval quality, staleness, and the propagation of a wrong prior remain open. The operator set Π and the external channel K are untouched: growing the operator library through multimodal distillation [6, 42, 73] and admitting external knowledge under a candidate-and-prune discipline [27] are future work.

10 Conclusion

Physical faithfulness in generated video requires three things that are usually studied apart: a way to *steer* what gets generated, a way to *measure* whether physical obligations are met, and *ground truth* against which measurements can be scored. This report contributes one artifact toward each. For steering, we build a

simulation-driven controllable generator (MuJoCo $\rightarrow$ Wan 2.2-VACE), a first step toward the control channel through which a critic might one day feed corrections back into generation. For ground truth, we assemble a benchmark of human-annotated, localized defects, so that a critic can be scored on whether it finds the specific fault rather than on aggregate preference. And for measurement, we present **PhyReAct**, an auditable multi-tool critic that grounds the reason-act-observe loop in physical verification: reasoning compiles the prompt into typed physical checks, frozen expert operators are called only to return measurements over the scope those checks declare, and a deterministic roll-up folds the observations into a three-valued verdict (**supported**/**contradicted**/**unknown**, surfaced as *plausible*, *implausible*, or **abstain**) with a human-traceable evidence chain.

On a development core of 149 clips carrying 304 human-marked defects, **PhyReAct** accounts for 228, against 164 for a published question-decomposition evaluator reading the same clips and checking the same claims with the same served model. Prompting the same backbone monolithically reaches 222, so the count alone does not separate the two. Every one of **PhyReAct**’s 228 arrives with the number that forced it, and a reader can audit any single verdict instead of accepting an aggregate.

That audibility is also what makes the evidence usable downstream. A contradicted claim names what failed, when it failed, and on what measurement, which is precisely the signal a generator needs to regenerate a clip locally rather than resample it whole. It is equally the substrate for an operator library that grows as new measurement procedures are distilled, and for improvement that edits a readable state rather than a weight. Closing the critic-to-generator loop over that signal is the next milestone.

A Appendix

To be expanded: the full generation configuration and control-render details; the benchmark schema and per-stratum statistics; the critic’s plan schema, predicate type rules, and temporal-relation truth tables; the specialist license and revision manifest; and the pre-registered evaluation manifest.

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